Investigating Oxidative Stress and Nrf2-Related Gene Expression Patterns in Noise-Induced Hearing Loss Using Bioinformatics and Cross-Dataset Validation

Final Biomedical Innovations Independent Research Paper

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ABSTRACT

Noise-induced hearing loss (NIHL) is a widespread form of sensorineural hearing loss that results from intense or prolonged noise exposure, but its biological damage extends beyond mechanical injury alone. Previous research suggests that oxidative stress, mitochondrial dysfunction, apoptosis, heat-shock response, and Nrf2-related antioxidant regulation all contribute to cochlear injury after noise exposure. The purpose of this study was to investigate whether oxidative stress and Nrf2-related gene expression patterns are associated with NIHL using an integrated bioinformatics approach. Publicly available gene expression datasets from the Gene Expression Omnibus were analyzed, with GSE100365 used as the primary discovery dataset and GSE12810 used as an independent validation dataset. Gene expression data were processed in Python, compared between control and noise-exposed groups, and analyzed using differential expression testing, annotation, principal component analysis, heatmap and volcano plot visualization, Random Forest machine learning, Gene Impact Score ranking, co-expression network analysis, and cross-dataset validation. In the discovery dataset, noise exposure was associated with transcriptomic changes involving mitochondrial regulation, cellular stress response, RNA processing, inflammatory signaling, and apoptosis-related pathways. In the validation dataset, strict genome-wide significance was limited by small sample size, but pathway-level validation supported a final oxidative stress/Nrf2-related gene set consisting of *Hspa1b, Jun, Nfe2l2, Dnm1l, Casp3, Bcl2, Dnaja1, Gclm, Nqo1, Sod2, Pink1,* and *Sirt3*. These results support the hypothesis that NIHL is associated with coordinated changes in oxidative stress and Nrf2-related gene expression rather than a single isolated gene. Genetic variation was part of the original conceptual framework, but because no direct SNP or GWAS analysis was completed, it is treated as a limitation and future direction. Overall, this project suggests that

bioinformatics and cross-dataset validation can help identify molecular stress-response patterns involved in NIHL susceptibility and may guide future research into protective or therapeutic targets.

BACKGROUND

Introduction

Noise-induced hearing loss (NIHL) is a widespread and preventable form of sensorineural hearing loss resulting from prolonged or intense exposure to environmental or occupational noise. It represents a significant global health burden, with millions of individuals at risk due to increasing exposure to recreational and industrial noise (World Health Organization [WHO], 2026). Although noise exposure is the primary environmental factor, individuals exposed to similar sound levels often exhibit markedly different degrees of hearing loss. This variability suggests that biological factors—particularly oxidative stress mechanisms, genetic variation, and gene regulatory networks—play a critical role in determining individual susceptibility.

Recent advances in auditory biology have demonstrated that NIHL is not caused by a single pathway, but instead results from interconnected molecular processes involving reactive oxygen species (ROS), mitochondrial dysfunction, inflammation, and dysregulation of antioxidant defense systems. In parallel, the increasing availability of large-scale genomic and transcriptomic datasets has enabled computational approaches to studying NIHL at a systems level. These approaches allow researchers to move beyond isolated gene studies and instead analyze gene networks, expression patterns, and predictive relationships across multiple datasets.

This literature review synthesizes current research on the molecular mechanisms of NIHL, the role of oxidative stress and mitochondrial dysfunction, genetic contributions to susceptibility, and the regulatory function of the Nrf2 pathway. It also highlights the growing importance of bioinformatics, network analysis, and machine learning approaches in identifying key genes and predictive patterns associated with NIHL.

Cellular and Molecular Mechanisms of Noise-Induced Hearing Loss

The cochlea is highly sensitive to mechanical and metabolic stress caused by excessive noise exposure. Damage occurs primarily in sensory hair cells and synaptic connections responsible for auditory signal transduction. Noise exposure induces both immediate mechanical injury and delayed biochemical damage, the latter being strongly associated with oxidative stress (Henderson et al., 2006; Kurabi et al., 2016).

A central feature of NIHL is the excessive production of reactive oxygen species (ROS), including superoxide radicals and hydroxyl species. Experimental studies demonstrate that ROS levels increase significantly following noise exposure, with both acute and delayed phases contributing to progressive cochlear injury (Yamane et al., 1995; Yamashita et al., 2004). These reactive molecules damage lipids, proteins, and DNA, ultimately impairing cellular integrity and triggering apoptotic pathways.

Mitochondria play a central role in this process as both a primary source and target of ROS. Noise-induced mitochondrial dysfunction reduces ATP production and further increases ROS generation, creating a self-amplifying feedback loop that accelerates cellular degeneration (Böttger & Schacht, 2013; Xu & Yang, 2024). This dysfunction activates apoptotic signaling pathways, including caspase-mediated cell death in cochlear hair cells (Hu et al., 2002; Nicotera et al., 2003).

Beyond hair cell loss, synaptic damage between inner hair cells and auditory nerve fibers—known as cochlear synaptopathy—can occur even in the absence of overt hair cell death. This contributes to “hidden hearing loss,” where auditory deficits are present despite normal audiometric thresholds (Liberman & Kujawa, 2017). Together, these findings demonstrate that NIHL is a multi-layered biological process involving mechanical injury, oxidative stress, mitochondrial dysfunction, and neural degeneration.

Oxidative Stress and Redox Homeostasis

Oxidative stress is widely recognized as a central driver of NIHL. Under normal physiological conditions, a balance exists between ROS production and antioxidant defense systems. However, noise exposure disrupts this balance, leading to redox homeostasis dysregulation and cellular injury (Fetoni et al., 2019; Zhou et al., 2023).

Endogenous antioxidant systems play a critical role in limiting oxidative damage. These include enzymatic defenses such as superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GPX), as well as non-enzymatic molecules such as glutathione. Experimental studies have shown that impairments in these systems significantly increase susceptibility to NIHL, indicating that antioxidant capacity is a key determinant of individual variability (McFadden et al., 2001; Ohlemiller et al., 2000).

Conversely, enhancement of antioxidant activity has been shown to provide protective effects. Glutathione supplementation reduces noise-induced damage, while combinations of antioxidant vitamins and magnesium attenuate cochlear injury and hearing threshold shifts (Ohinata et al., 2000; Prell et al., 2007). Mitochondria-targeted antioxidants have also demonstrated protective effects by directly reducing mitochondrial ROS production and preserving cellular function (Kim et al., 2018; Baek et al., 2023).

Importantly, oxidative stress also interacts with other biological processes, including inflammation, autophagy, and activation of the NLRP3 inflammasome, further contributing to cochlear degeneration. These interconnected processes highlight the need for systems-level approaches to understanding NIHL rather than isolated single-pathway analysis.

Genetic Contributions to Susceptibility

A substantial body of evidence demonstrates that genetic variation plays a major role in determining susceptibility to NIHL. Despite similar environmental exposure, individuals exhibit wide variation in hearing outcomes, suggesting that genetic differences modulate oxidative stress responses and cellular resilience.

Polymorphisms in genes involved in antioxidant defense, including SOD, CAT, and paraoxonase genes, have been associated with increased risk of hearing loss (Fortunato et al., 2004; Carlsson et al., 2004). These variants may alter enzymatic efficiency, reducing the ability to neutralize ROS and increasing vulnerability to cochlear damage.

Genome-wide association studies (GWAS) have further identified multiple genetic loci associated with hearing function and susceptibility, indicating that NIHL is a polygenic trait influenced by multiple interacting biological pathways (Girotto et al., 2011; Vuckovic et al., 2015). Similarly, animal models have demonstrated that genetic background significantly affects susceptibility to noise-induced damage, reinforcing the role of heritable variation in auditory resilience (Davis et al., 2001).

However, while many genetic associations have been identified, the functional relationships between gene variants, expression patterns, and biological pathways remain insufficiently understood. This gap highlights the need for integrative computational approaches that combine genetic, transcriptomic, and pathway-level data.

The Nrf2 Pathway and Antioxidant Regulation

The Nrf2 (nuclear factor erythroid 2–related factor 2) signaling pathway is a key regulator of cellular antioxidant responses. Nrf2 controls the expression of multiple detoxification and antioxidant genes, thereby playing a central role in protecting cells from oxidative stress (Li et al., 2021; Abdolmaleki et al., 2024).

Experimental studies have demonstrated that activation of the Nrf2 pathway reduces oxidative damage in cochlear cells and protects against noise-induced hearing loss. Nrf2 activation enhances expression of cytoprotective genes and preserves hair cell integrity following noise exposure (Honkura et al., 2016). Additionally, compounds that activate the Nrf2/HO-1 pathway have shown significant protective effects in experimental models of cochlear injury.

Recent research has also highlighted interactions between Nrf2 and other regulatory pathways, including SIRT1, which together modulate mitochondrial function and oxidative stress responses (Zheng et al., 2024; Luo et al., 2025). These findings suggest that NIHL is regulated by interconnected signaling networks rather than isolated pathways.

Despite these findings, the precise role of Nrf2 within broader gene regulatory networks remains unclear, particularly in terms of how gene expression variability and genetic variation influence its protective effects across individuals.

Computational Approaches in NIHL Research

Recent advances in bioinformatics have enabled large-scale analysis of gene expression and genetic variation data relevant to hearing loss. Publicly available datasets, such as those from the Gene Expression Omnibus, allow researchers to investigate transcriptional changes across conditions and identify genes associated with oxidative stress responses.

Similarly, genomic variation data from Ensembl provide insights into how genetic differences may influence gene function and susceptibility to disease. Pathway databases such as KEGG allow for the mapping of gene interactions and regulatory networks.

In addition to traditional statistical approaches, machine learning methods are increasingly being applied in biomedical research to analyze high-dimensional gene expression data. These methods can identify complex nonlinear relationships between genes and classify biological states based on expression patterns. In the context of NIHL, machine learning provides a powerful tool for identifying predictive gene signatures associated with susceptibility.

Network-based approaches further enhance this analysis by modeling gene-gene interactions and identifying central regulatory genes within biological pathways. Together, these computational approaches enable a systems-level understanding of oxidative stress responses in hearing loss that is not possible through single-gene analysis alone.

Gaps in Current Knowledge

Despite significant advances, several key gaps remain in the understanding of NIHL. First, while oxidative stress is well established as a central mechanism, the integrated relationships between mitochondrial dysfunction, inflammation, and antioxidant defenses remain incompletely characterized.

Second, although genetic associations have been identified, the functional mechanisms linking genetic variation to gene expression patterns and pathway activity are not fully understood.

Third, while antioxidant therapies have shown protective effects in experimental models, their effectiveness across different biological contexts remains inconsistent, limiting their clinical translation.

Finally, there is a lack of integrative computational studies that combine gene expression data, genetic variation, pathway information, and machine learning-based predictive modeling to identify key regulatory genes and susceptibility patterns in NIHL. This highlights the need for data-driven, systems-level approaches.

Although this literature review originally framed NIHL susceptibility through oxidative stress, Nrf2 regulation, gene expression, and genetic variation, the completed experimental phase of this project focused specifically on transcriptomic gene expression analysis. Therefore, genetic variation remains important background information because it helps explain why individuals may respond differently to similar noise exposure, but it was not directly tested in the completed analysis. Instead, this study used two public gene expression datasets to investigate whether oxidative stress and Nrf2-related expression patterns were associated with noise exposure. GSE100365 served as the discovery dataset, while GSE12810 served as an independent validation dataset. This adjustment keeps the biological purpose of the literature review intact while aligning the final paper with the data that were actually analyzed.

Research Question

Based on these gaps, the research question guiding this study is:

- How do gene expression patterns, genetic variation, and gene network interactions within oxidative stress-related pathways—particularly the Nrf2 signaling pathway—predict susceptibility to noise-induced hearing loss using integrated bioinformatics and machine learning approaches?

HYPOTHESIS AND PURPOSE

The purpose of this study was to investigate whether oxidative stress and Nrf2-related gene expression patterns are associated with noise-induced hearing loss using publicly available transcriptomic datasets. This project used GSE100365 as the primary discovery dataset and GSE12810 as an independent validation dataset. Instead of focusing on a single gene, this study examined whether noise exposure produced broader gene expression changes connected to oxidative stress, mitochondrial dysfunction, apoptosis, heat-shock response, inflammatory signaling, and antioxidant defense.

The original research question guiding this study was: **How do gene expression patterns, genetic variation, and gene network interactions within oxidative stress-related pathways, particularly the Nrf2 signaling pathway, predict susceptibility to noise-induced hearing loss using integrated bioinformatics and machine learning approaches?** However, after the experimental phase was completed, the final analysis focused specifically on gene expression patterns and cross-dataset validation. Genetic variation remained biologically relevant to the overall topic, but it was not directly analyzed through SNP, GWAS, or variant-level methods. Therefore, genetic variation is treated in this final paper as a limitation and future direction rather than as a completed result.

The hypothesis for this study was: **If cochlear or auditory nerve tissue is exposed to noise, then oxidative stress and Nrf2-related genes will show altered expression compared with**

control tissue, because noise-induced hearing loss activates cellular stress, mitochondrial damage, apoptosis, and antioxidant defense pathways.

In this hypothesis, the independent variable was **noise exposure condition**, comparing control tissue to noise-exposed tissue. The dependent variable was **gene expression activity**, especially the expression of genes related to oxidative stress, Nrf2 signaling, mitochondrial regulation, apoptosis, and cellular stress response. The expected outcome was that noise-exposed samples would show measurable transcriptomic differences from control samples and that some of these differences would be supported across both the discovery and validation datasets.

MATERIALS/METHODS

Materials

This project used publicly available gene expression datasets from the National Center for Biotechnology Information Gene Expression Omnibus (NCBI GEO) to investigate oxidative stress and Nrf2-related gene expression patterns in noise-induced hearing loss. The primary discovery dataset was **GSE100365**, and the independent validation dataset was **GSE12810**. Both datasets used the **GPL1261 Affymetrix Mouse Genome 430 2.0 Array** platform, which made cross-dataset comparison more consistent because both datasets measured gene expression using the same microarray platform.

The main materials and tools used in this project included a computer with internet access, Jupyter Notebook, Python, and publicly available bioinformatics resources. Python libraries used during analysis included **pandas** and **NumPy** for data cleaning and organization, **SciPy** for statistical testing, **statsmodels** for multiple-testing correction, **scikit-learn** for principal component analysis and Random Forest machine learning, **matplotlib** and **seaborn** for visualization, and **GEOparse** for downloading and processing the GSE12810 validation dataset. Platform annotation information from GPL1261 was used to connect microarray probe IDs to gene symbols, gene titles, Entrez gene IDs, and Gene Ontology information.

The project also used background research from the literature review to interpret genes and pathways related to oxidative stress, reactive oxygen species, mitochondrial dysfunction, apoptosis, heat-shock response, inflammatory signaling, antioxidant defense, and Nrf2 pathway regulation. Genetic variation was included in the original background and project framing, but no direct SNP, GWAS, or variant-level analysis was completed. Therefore, genetic variation was treated as a limitation and future direction rather than as a completed analysis.

Figures during the project included PCA plots, heatmaps, volcano plots, Random Forest outputs, feature-importance tables, Gene Impact Score rankings, co-expression network diagrams, annotated differential expression tables, oxidative stress/Nrf2 candidate gene tables, cross-dataset validation tables, and final validated gene-set summaries. The strongest figures and tables are included in the Data section.

Methods

This project used an integrated transcriptomic bioinformatics workflow to analyze gene expression patterns associated with noise-induced hearing loss. The analysis was completed in two major phases. First, GSE100365 was used as the discovery dataset to identify candidate genes, expression patterns, and biological pathways associated with noise exposure. Second, GSE12810 was used as an independent validation dataset to test whether oxidative stress and Nrf2-related expression patterns were supported across datasets.

Dataset Selection

GSE100365 was selected as the primary discovery dataset because it included mouse auditory nerve samples from control and noise-exposed conditions. After preprocessing, this dataset contained **15 total samples**, including **3 control samples** and **12 noise-exposed samples**. The noise-exposed group included multiple post-noise timepoints: immediate, 1 day, 7 days, and 14 days after noise exposure. For the main differential expression and machine learning analysis, these timepoints were collapsed into a binary comparison of control versus noise-exposed samples. However, timepoint-based PCA was also used to examine whether gene expression patterns changed over time after noise exposure.

GSE12810 was selected as the independent validation dataset. This dataset contained mouse cochlear modiolus or spiral ganglion-related samples from noise-exposed and negative-control conditions. It contained **6 total samples**, including **3 noise-exposed samples** and **3 control samples**. GSE12810 was especially useful for validation because it used the same GPL1261 Affymetrix Mouse Genome 430 2.0 Array platform as GSE100365. The validation dataset was not treated as a full replacement for the discovery dataset; instead, it was used to test whether the main oxidative stress and Nrf2-related patterns from GSE100365 showed support in an independent dataset.

Data Download and Preprocessing

For GSE100365, the GEO series matrix file was downloaded and imported into Python using pandas. Metadata lines beginning with an exclamation point were excluded so that the gene expression matrix could be read correctly. Probe IDs were preserved, and the expression matrix was transposed so that each row represented a sample and each column represented a probe. Expression values were converted to numeric values, and sample labels were extracted from the sample title metadata. The final GSE100365 expression matrix contained **15 samples and 45,101 probes**, with no missing expression values detected.

For GSE12810, the dataset was downloaded using GEOparse. Expression values from each sample were extracted and merged by probe ID to create one complete expression matrix. This matrix contained **45,101 probes and 6 samples**. The expression values ranged from approximately 2 to 14, which indicated that the values were already log2-normalized microarray expression values. Because of this, a second log2 transformation was not applied.

Group Labeling

Samples were assigned to biological groups based on their GEO sample titles. In GSE100365, samples containing "control" in the title were labeled as control, while samples containing noise-exposure labels were labeled as noise-exposed. This produced a binary grouping where control samples were coded as 0 and noise-exposed samples were coded as 1.

In GSE12810, samples labeled as noise-exposed modioli were assigned to the noise group, while samples labeled as negative-control modioli were assigned to the control group. These group labels were used for differential expression analysis, PCA visualization, heatmap organization, and cross-dataset validation.

Differential Expression Analysis

Differential expression analysis was performed to compare gene expression between control and noise-exposed samples. For each probe, the mean expression value was calculated for the control group and the noise-exposed group. A **Welch's t-test** was used because it does not assume equal variance between groups and is more appropriate when sample sizes are unequal, especially for GSE100365, which contained 3 controls and 12 noise-exposed samples.

For each probe, the following values were calculated:

- control mean expression
- noise-exposed mean expression
- t-statistic
- p-value
- approximate log2 fold change
- absolute log2 fold change

Because the microarray expression values were already log2-like, the difference between the noise-exposed mean and the control mean was treated as an approximate log2 fold change. During statistical testing, precision-loss warnings occurred for some probes because certain expression values were nearly identical across samples. These warnings were noted as a limitation because they indicated that some individual t-test estimates may be unstable.

For GSE12810, p-values were also adjusted using the **Benjamini-Hochberg false discovery rate correction** because thousands of probes were tested at the same time. Under strict criteria using adjusted p-value and fold-change thresholds, GSE12810 produced **0 significant probes**. This was not treated as a failure of the project because the validation dataset had only 6 samples and was underpowered for strict genome-wide discovery. Instead, GSE12810 was used for ranked validation, candidate-gene checking, and pathway-level interpretation.

Probe Annotation

The GPL1261 platform annotation file was used to connect microarray probe IDs to interpretable biological information. This step was necessary because the original expression matrices were organized by probe ID rather than by gene name. Probe annotation added gene symbols, gene titles, Entrez gene IDs, Gene Ontology biological processes, Gene Ontology molecular functions, and Gene Ontology cellular components.

This annotation step allowed the statistical output to be interpreted biologically. For example, top-ranked probes could be connected to genes involved in mitochondrial dynamics, heat-shock response, RNA processing, apoptosis, inflammatory signaling, antioxidant defense, and Nrf2-related stress response.

Exploratory Data Analysis and Visualization

Several exploratory visualizations were generated to examine the structure of the datasets and to communicate the major expression patterns. For GSE100365, principal component analysis was performed using the full expression matrix to reduce the high-dimensional gene expression data into two principal components. A binary PCA plot was generated to compare control and noise-exposed samples, and a timepoint PCA plot was generated to compare control, immediate, 1-day, 7-day, and 14-day post-noise samples.

A z-score heatmap was generated using the top differentially expressed genes or probes from GSE100365. Z-score standardization allowed relative high and low expression patterns to be compared across samples. A volcano plot was also generated to show the relationship between approximate log2 fold change and statistical significance.

For GSE12810, a sample correlation heatmap and PCA plot were generated to evaluate sample quality and determine whether control and noise-exposed samples showed separation. Additional validation figures were created, including an oxidative stress/Nrf2 heatmap, a GSE12810 volcano plot, a top ranked DEG heatmap, cross-dataset log2 fold-change scatterplots, a final validated gene heatmap, and a final validated gene log2 fold-change comparison plot.

Machine Learning Analysis

A Random Forest classifier was trained using the GSE100365 discovery dataset to test whether gene expression profiles could distinguish control samples from noise-exposed samples. The model used **500 trees**, balanced class weights, and 3-fold stratified cross-validation. The initial cross-validation accuracy was approximately **0.80**, but this value was interpreted cautiously because the dataset was imbalanced, with only 3 control samples and 12 noise-exposed samples.

Further evaluation showed that the model mainly predicted the majority class, meaning that the raw accuracy overstated the model's true performance. Because of this, the Random Forest model was not treated as a reliable diagnostic or predictive classifier. Instead, it was used as an exploratory feature-prioritization tool. Random Forest feature importance helped identify genes that may contribute to expression-based differences between control and noise-exposed samples.

Gene Impact Score Development

A custom **Gene Impact Score** was developed to prioritize genes using multiple layers of evidence rather than relying only on p-values. The first version of the Gene Impact Score combined statistical significance, fold-change magnitude, machine learning feature importance, and biological relevance. A second version added network centrality, making it a stronger integrated ranking system.

The Gene Impact Score was designed to highlight genes that were important across several dimensions of the project. Genes with higher scores were considered stronger candidates because they showed evidence from differential expression, machine learning, biological interpretation, and/or network structure. This method helped identify genes related to mitochondrial regulation, heat-shock response, RNA processing, inflammation, apoptosis, and oxidative stress.

Co-Expression Network Analysis

A co-expression network was created using top-ranked candidate genes from the Gene Impact Score analysis. In the network, each gene was represented as a node, and strong correlations between gene expression patterns were represented as edges. A correlation threshold of approximately **0.70** was used to identify strong co-expression relationships.

The resulting network contained **22 nodes** and **48 edges**. This network was interpreted as exploratory because the dataset was small, but it still helped show that the candidate genes were not acting as isolated markers. Instead, the results suggested coordinated gene expression patterns involving stress response, mitochondrial regulation, RNA processing, and cellular injury pathways.

Cross-Dataset Validation

After the discovery analysis was completed using GSE100365, GSE12810 was used to validate the major candidate genes and pathway-level patterns. First, the major Dataset 1 candidate genes were searched in GSE12810. These included *Hspa1b, Dnm1l, Stat1, Gls, Dnaja2, Hnrnpu, Pde3b,* and *Srsf1*. All eight were found in GSE12810, although not all showed strong validation trends.

Next, a targeted oxidative stress/Nrf2-related gene list was tested in GSE12810. This list included genes involved in core Nrf2 signaling, antioxidant defense, ROS detoxification, mitochondrial stress, apoptosis, inflammation, and heat-shock response. GSE12810 produced stronger support at the pathway level than at the individual-gene level. Therefore, validation was interpreted as evidence for a broader oxidative stress/Nrf2-related response rather than perfect replication of every individual gene from the discovery dataset.

Cross-dataset comparisons were also performed by comparing log2 fold-change direction and magnitude between GSE100365 and GSE12810. A whole-transcriptome comparison was generated for overlapping genes, and a more focused oxidative stress/Nrf2 comparison was generated for overlapping genes in the targeted pathway set. Direction agreement was used as one piece of evidence in deciding which genes were most consistently supported across datasets.

Final Validated Gene Set Selection

The final validated oxidative stress/Nrf2-related gene set was selected by combining discovery evidence from GSE100365, validation evidence from GSE12810, cross-dataset comparison, and biological relevance from the literature review. The final validated gene set consisted of:
Hspa1b, Jun, Nfe2l2, Dnm1l, Casp3, Bcl2, Dnaja1, Gclm, Nqo1, Sod2, Pink1, and *Sirt3*.

These genes were selected because they represented major biological processes connected to NIHL, including heat-shock response, transcriptional stress signaling, Nrf2 regulation, mitochondrial dynamics, apoptosis, antioxidant defense, mitochondrial quality control, and redox balance.

DATA

This project produced gene expression data, statistical results, machine learning outputs, visualizations, and cross-dataset validation summaries from two public GEO datasets. GSE100365 was used as the primary discovery dataset, while GSE12810 was used as the independent validation dataset. The purpose of this section is to present the major results generated during the computational analysis before interpreting their biological meaning in the Analysis section.

Dataset Overview

Two datasets were used in this project. Dataset 1, GSE100365, served as the discovery dataset and contained mouse auditory nerve or cochlear-associated tissue samples from control and noise-exposed conditions. Dataset 2, GSE12810, served as the validation dataset and contained mouse cochlear modiolus or spiral ganglion-related samples from noise-exposed and negative-control conditions. Both datasets used the GPL1261 Affymetrix Mouse Genome 430 2.0 Array platform, which made cross-dataset comparison more technically consistent.

Table 1. Summary of datasets used in the project.

Dataset	GEO Accession	Role	Samples	Groups	Platform	Main Use
Dataset 1	GSE100365	Discovery dataset	15	3 control, 12 noise-exposed	GPL1261 Affymetrix Mouse Genome 430 2.0 Array	Differential expression, PCA, heatmap, volcano plot, machine learning, Gene Impact Score, co-expression network
Dataset 2	GSE12810	Validation dataset	6	3 control, 3 noise-exposed	GPL1261 Affymetrix Mouse Genome 430 2.0 Array	Candidate validation, oxidative stress/Nrf2 validation, cross-dataset comparison

GSE100365 contained 45,101 probes after preprocessing. The final expression matrix was organized as 15 samples by 45,101 probes, with no missing expression values detected. GSE12810 also contained 45,101 probes across 6 samples. Because both datasets used the same GPL1261 platform, the analysis could compare shared genes and oxidative stress/Nrf2-related patterns across datasets.

Dataset 1: GSE100365 Discovery Results

GSE100365 was used to identify the major gene expression changes associated with noise exposure. The dataset included 3 control samples and 12 noise-exposed samples. The noise-exposed samples represented multiple post-noise timepoints: immediate, 1 day, 7 days, and 14 days. For the main analysis, the timepoints were collapsed into one noise-exposed group, but timepoint-specific visualization was also used to examine whether gene expression patterns changed across the post-noise recovery period.

Table 2. GSE100365 sample structure.

Group	Number of Samples	Description
Control	3	Auditory nerve control samples
Noise-exposed	12	Auditory nerve samples collected immediately, 1 day, 7 days, and 14 days after noise exposure
Total	15	Full discovery dataset

Principal component analysis was used to visualize overall gene expression patterns in GSE100365. The binary PCA plot compared control samples against all noise-exposed samples. PC1 explained approximately 19.2% of the variance, while PC2 explained approximately 15.0% of the variance. Together, the first two principal components explained about 34% of the total variation in the dataset. The binary PCA plot showed partial separation between control and noise-exposed samples, suggesting that noise exposure was associated with broad transcriptomic differences.

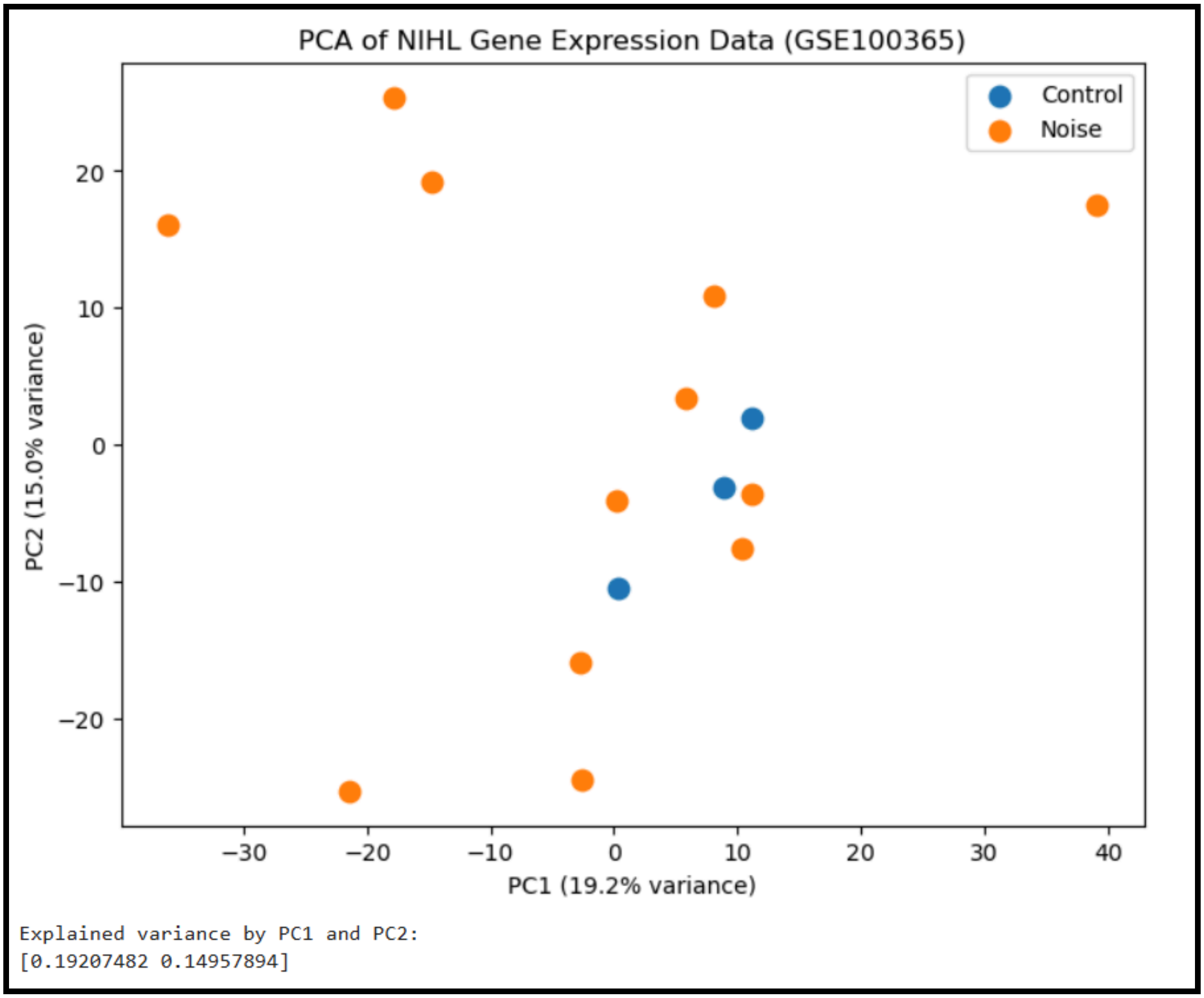


Figure 1. Binary PCA plot of GSE100365 control and noise-exposed samples.

This PCA plot shows partial separation between control and noise-exposed samples based on whole-transcriptome gene expression.

A second PCA plot separated the samples by post-noise timepoint. This figure was more biologically informative than the binary PCA because it showed that the noise-exposed samples did not behave as one uniform group. Instead, samples from different timepoints occupied different regions of the PCA space. This suggests that gene expression responses after acoustic trauma may shift over time through different stages of injury, stress response, repair, or degeneration. In the timepoint PCA, PC1 explained 19.2% of the variance and PC2 explained 15.0%, for a combined 34.2% of total variance.

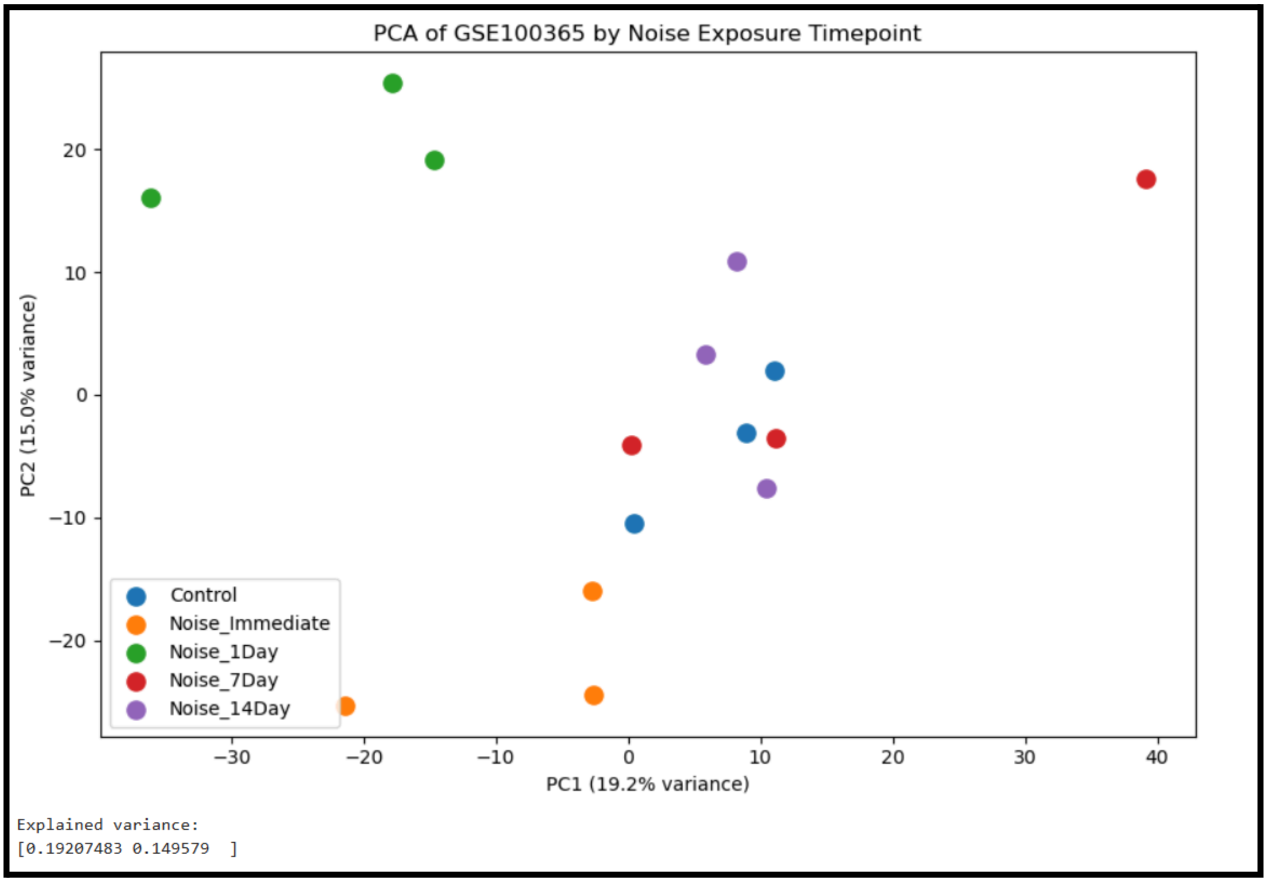


Figure 2. Timepoint PCA plot of GSE100365 samples.

This plot shows that post-noise timepoints displayed different expression patterns, supporting the idea that the molecular response to noise exposure changes over time.

Differential expression analysis was performed using Welch's t-test to compare control and noise-exposed samples. Several top-ranked genes were connected to RNA regulation, mitochondrial dynamics, heat-shock response, inflammatory signaling, cytoskeletal regulation, and cellular stress response. Important discovery dataset candidates included *Hnrnpu, Dnm1l, Dnaja2, Stat1, Hspa1b, Pde3b,* and *Srsf1*. The GSE100365 pipeline identified *Hnrnpu* and

Dnm1l among the top annotated differential expression results, with *Dnm1l* especially relevant because of its connection to mitochondrial dynamics.

Table 3. Top annotated differentially expressed genes in GSE100365.

Gene	**Main Biological Relevance**
Hnrnpu	RNA processing, nuclear regulation, mRNA stabilization
Dnm1l	Mitochondrial fission, mitochondrial stress, ROS-linked dysfunction
Dnaja2	Heat-shock response, chaperone activity, protein-folding stress
Stat1	Inflammatory signaling and stress response
Hspa1b	Heat-shock response and cellular stress adaptation
Pde3b	Cellular signaling
Srsf1	RNA splicing and post-transcriptional regulation

A z-score heatmap was generated using the top 25 differentially expressed genes or probes from GSE100365. The heatmap showed relative expression patterns across samples. Control samples appeared more consistent, while noise-exposed samples showed broader variation across the selected genes. This supported the idea that noise exposure was associated with transcriptomic remodeling in auditory nerve tissue.

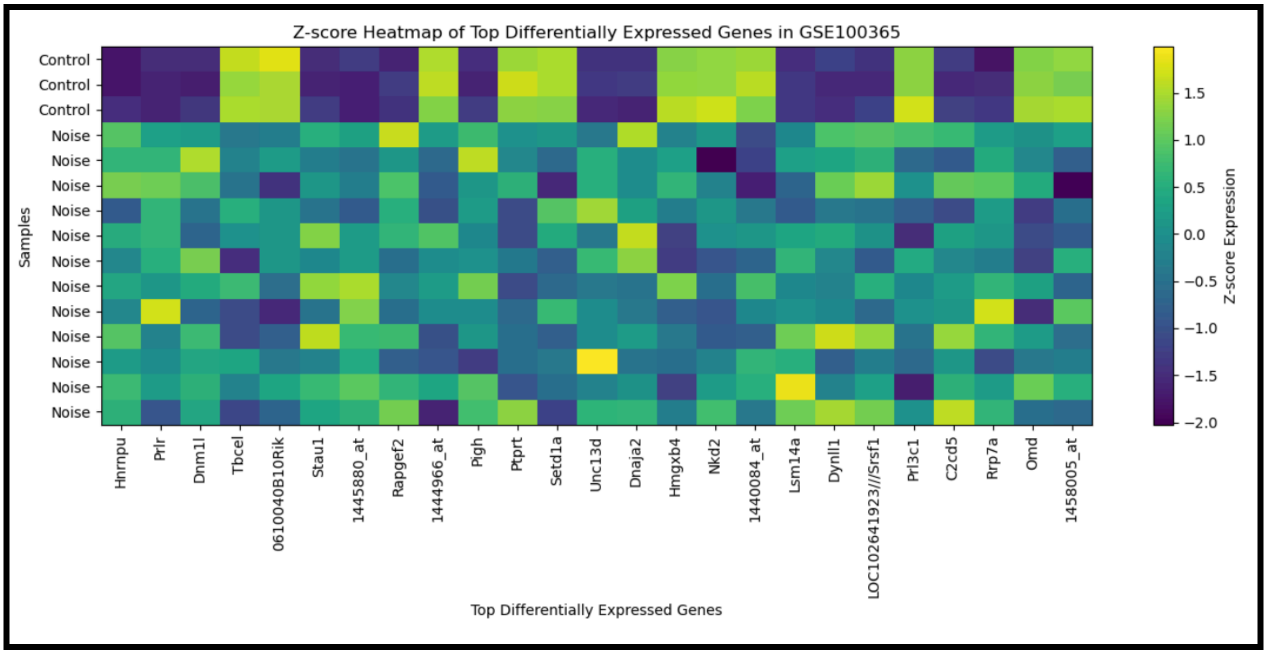


Figure 3. Z-score heatmap of top differentially expressed genes in GSE100365.

This heatmap shows relative expression differences across control and noise-exposed samples for the top-ranked genes.

A volcano plot was generated to show the relationship between approximate log2 fold change and statistical significance. The plot helped identify genes that had both meaningful expression changes and relatively strong p-values. Several genes of interest were highlighted, including *Hspa1b, Dnm1l, Stat1, Gls, Sod1,* and *Dnaja2*.

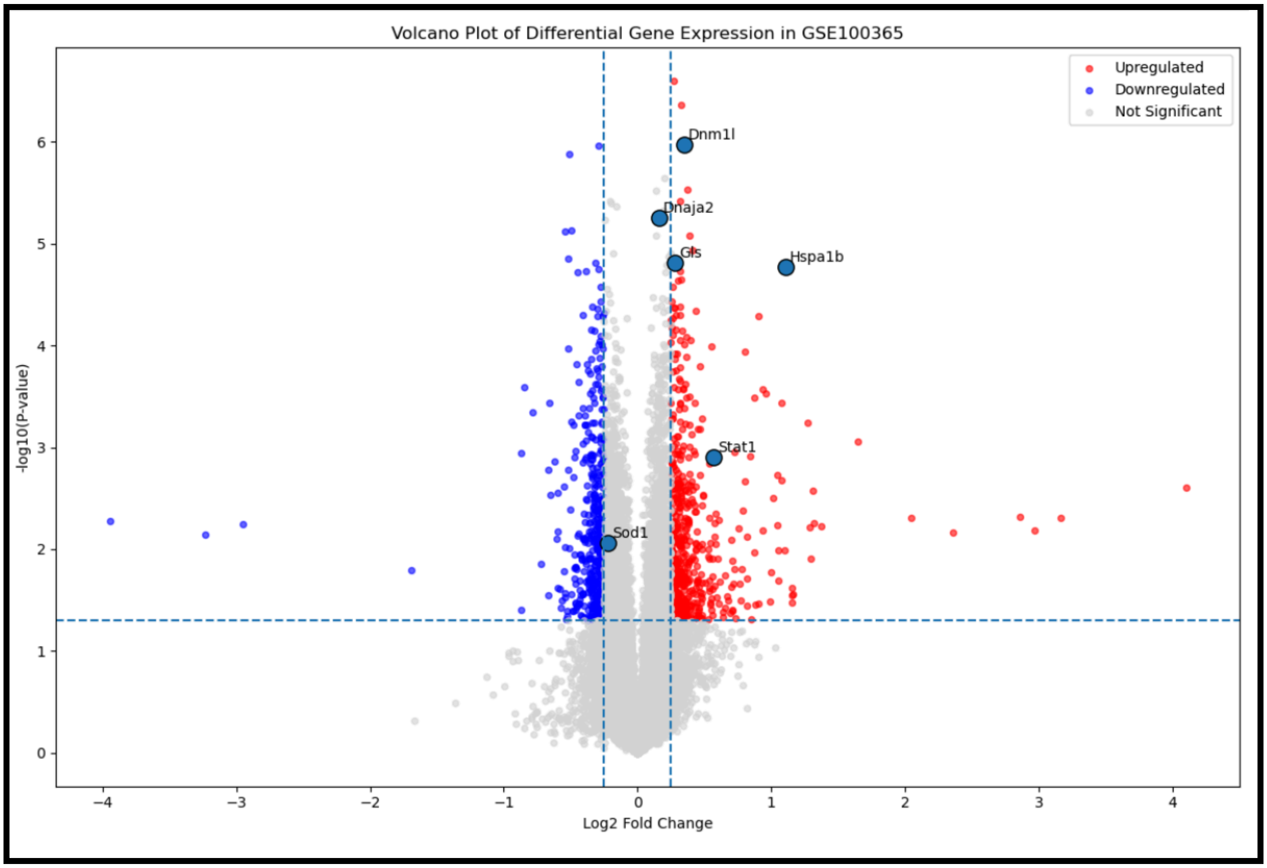


Figure 4. Volcano plot of GSE100365 differential expression results.

This plot shows differential expression trends by comparing approximate log2 fold change with statistical significance.

A targeted oxidative stress/Nrf2 candidate analysis was also performed in GSE100365. This analysis focused on genes related to antioxidant defense, Nrf2 signaling, mitochondrial stress, ROS detoxification, apoptosis, heat-shock response, and inflammatory signaling. The strongest signals were not limited to classic antioxidant enzymes. Instead, *Hspa1b, Dnm1l, Stat1,* and *Gls* helped connect the discovery dataset to broader stress-response and mitochondrial injury pathways.

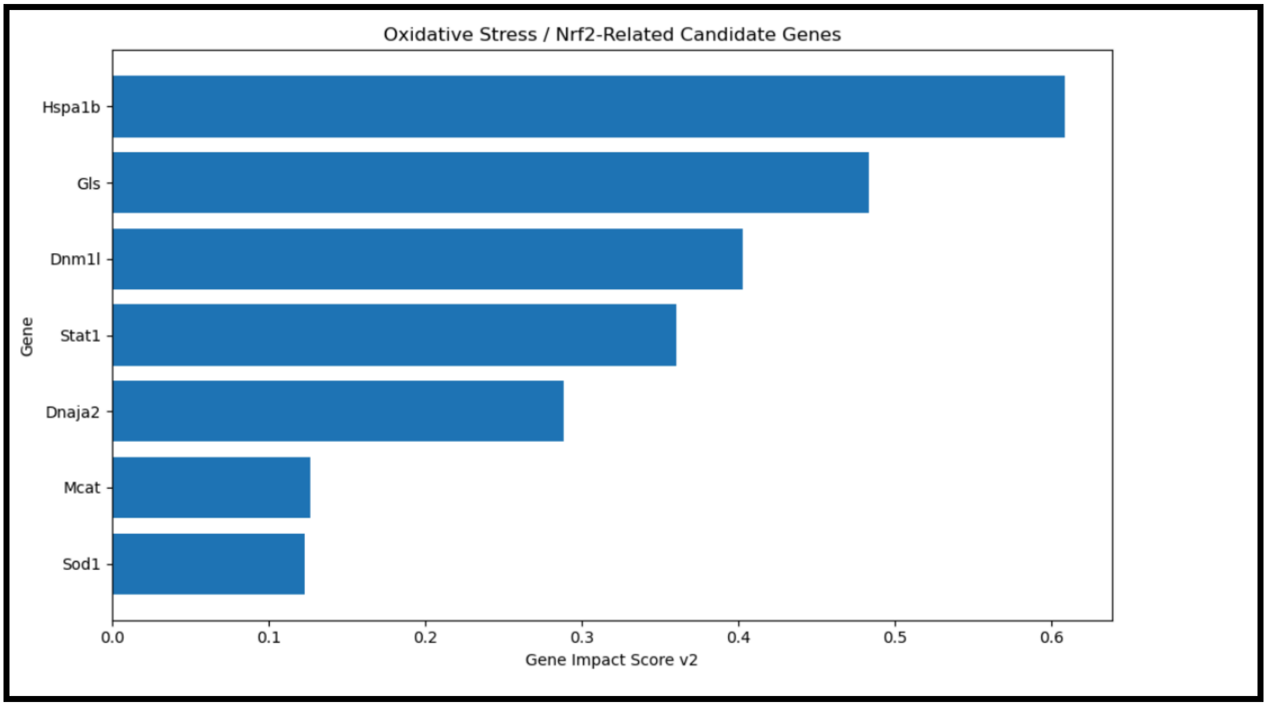


Figure 5. Oxidative stress/Nrf2 candidate gene expression trends in GSE100365.

This figure highlights selected genes connected to oxidative stress, mitochondrial dysfunction, heat-shock response, inflammatory signaling, and Nrf2-related pathways.

A Random Forest classifier was trained using the GSE100365 expression matrix to test whether gene expression profiles could classify samples as control or noise-exposed. The initial 3-fold cross-validation accuracy was approximately 0.80. However, because the dataset was imbalanced, with 12 noise-exposed samples and only 3 control samples, the accuracy was misleading. Further evaluation showed that the model mainly predicted the majority class. Therefore, the Random Forest model was not treated as a reliable predictive classifier, but it was still useful for exploratory feature prioritization.

```
Confusion Matrix:
[[ 0  3]
 [ 0 12]]

Classification Report:
              precision    recall  f1-score   support

     Control       0.00      0.00      0.00         3
       Noise       0.80      1.00      0.89        12

    accuracy                           0.80        15
   macro avg       0.40      0.50      0.44        15
weighted avg       0.64      0.80      0.71        15


Balanced Accuracy:
0.5
```

Supplemental Table/Figure S1. Random Forest model evaluation for GSE100365.

This output shows that the model's raw accuracy was limited by class imbalance and should be interpreted cautiously.

Random Forest feature importance identified several genes that may contribute to expression-based differences between control and noise-exposed samples. Important feature-importance genes included *Hspa1b, Stat1, Pde3b, Dnajb4, Arid5b, Rab11fip4,* and *Nfic*. These genes were connected to cellular stress response, inflammation, signal transduction, chaperone activity, transcriptional regulation, and vesicular trafficking.

Table 4. Selected Random Forest feature-importance genes from GSE100365.

Gene	**Biological Relevance**
Hspa1b	Heat-shock and cellular stress response
Stat1	Inflammatory signaling

Pde3b	Signal transduction
Dnajb4	Chaperone and stress response
Arid5b	Transcription regulation
Rab11fip4	Vesicular trafficking
Nfic	Transcription factor activity

A custom Gene Impact Score was developed to prioritize genes using multiple types of evidence. Gene Impact Score v2 combined differential expression, fold-change magnitude, machine learning feature importance, biological relevance, and network centrality. This produced an integrated ranking of candidate genes rather than relying only on p-values. In the updated Gene Impact Score, *Hnrnpu* became the highest-ranked candidate after network centrality was included, while *Hspa1b, Dnm1l, Stat1, Dnaja2, Pde3b,* and *Srsf1* remained important candidates.

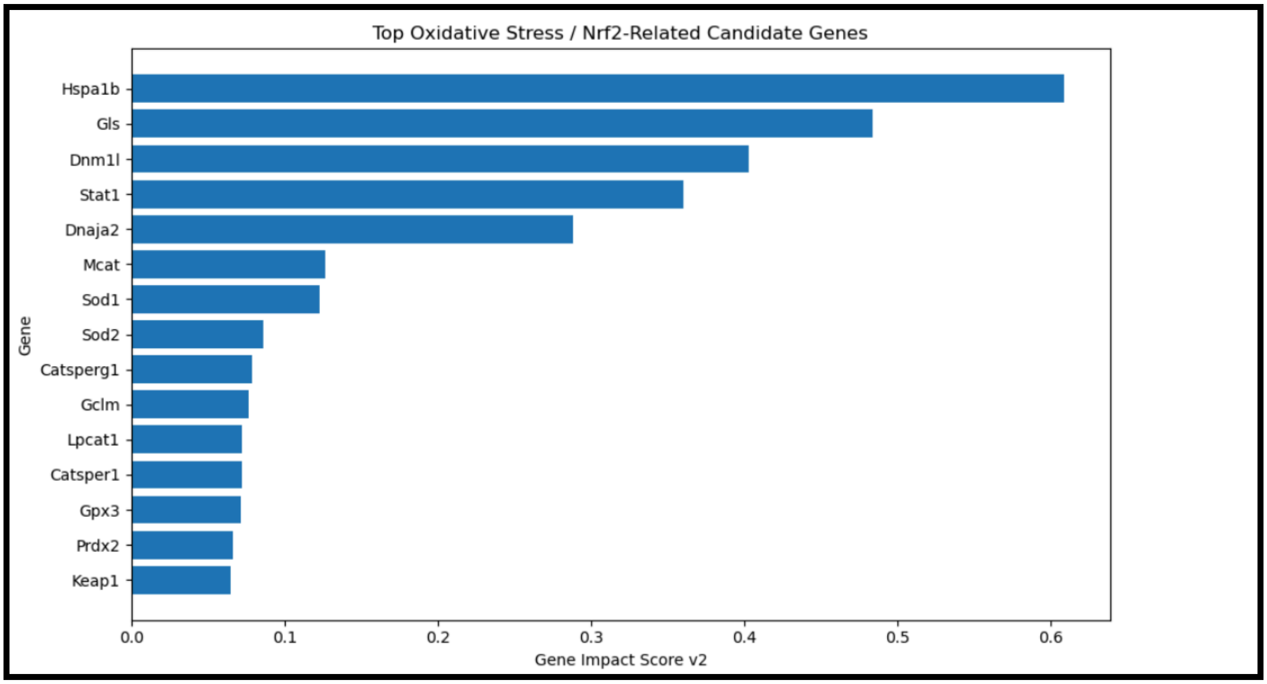


Figure 6. Gene Impact Score v2 ranking of top candidate genes.

This figure shows integrated candidate gene rankings based on statistical, machine learning, biological, and network-based evidence.

Table 5. Major Gene Impact Score v2 candidates from GSE100365.

Gene	**Main Interpretation**
Hnrnpu	RNA regulation and high network centrality
Hspa1b	Heat-shock response
Dnm1l	Mitochondrial dysfunction and mitochondrial fission
Stat1	Stress and inflammatory signaling
Dnaja2	Cellular stress adaptation

Pde3b	Cellular signaling
Srsf1	RNA processing

A co-expression network was generated using top-ranked candidate genes. The network contained 22 nodes and 48 edges, indicating that several candidate genes showed coordinated expression patterns. This network was interpreted as exploratory because the dataset was small, but it supported the idea that NIHL-related gene expression changes occur in connected biological modules rather than as isolated single-gene events.

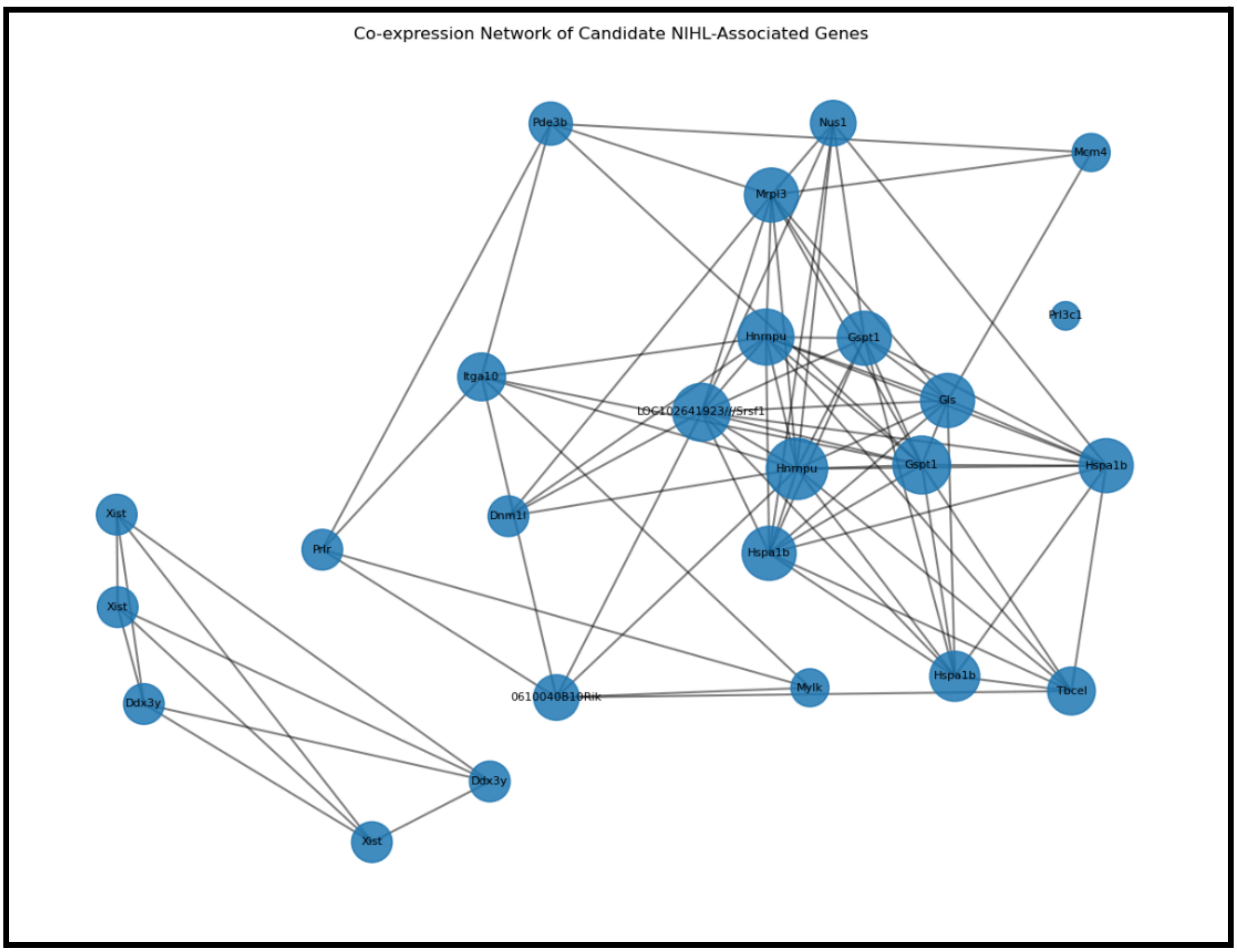


Figure 7. Co-expression network of top GSE100365 candidate genes.

This exploratory network shows relationships between candidate genes based on correlated expression patterns.

Dataset 2: GSE12810 Validation Results

GSE12810 was used as an independent validation dataset. This dataset contained 6 total samples: 3 noise-exposed samples and 3 negative-control samples. The validation dataset used the same GPL1261 microarray platform as GSE100365, which allowed gene-level comparison across datasets.

Table 6. GSE12810 sample structure.

Group	Sample IDs	Number of Samples
Noise-exposed	GSM321743, GSM321744, GSM321745	3
Control	GSM321746, GSM321747, GSM321748	3
Total	All samples	6

A sample correlation heatmap was generated for GSE12810. The samples were highly correlated overall, with correlations around 0.98 to 1.00. This suggested that there were no major failed samples or catastrophic outliers.

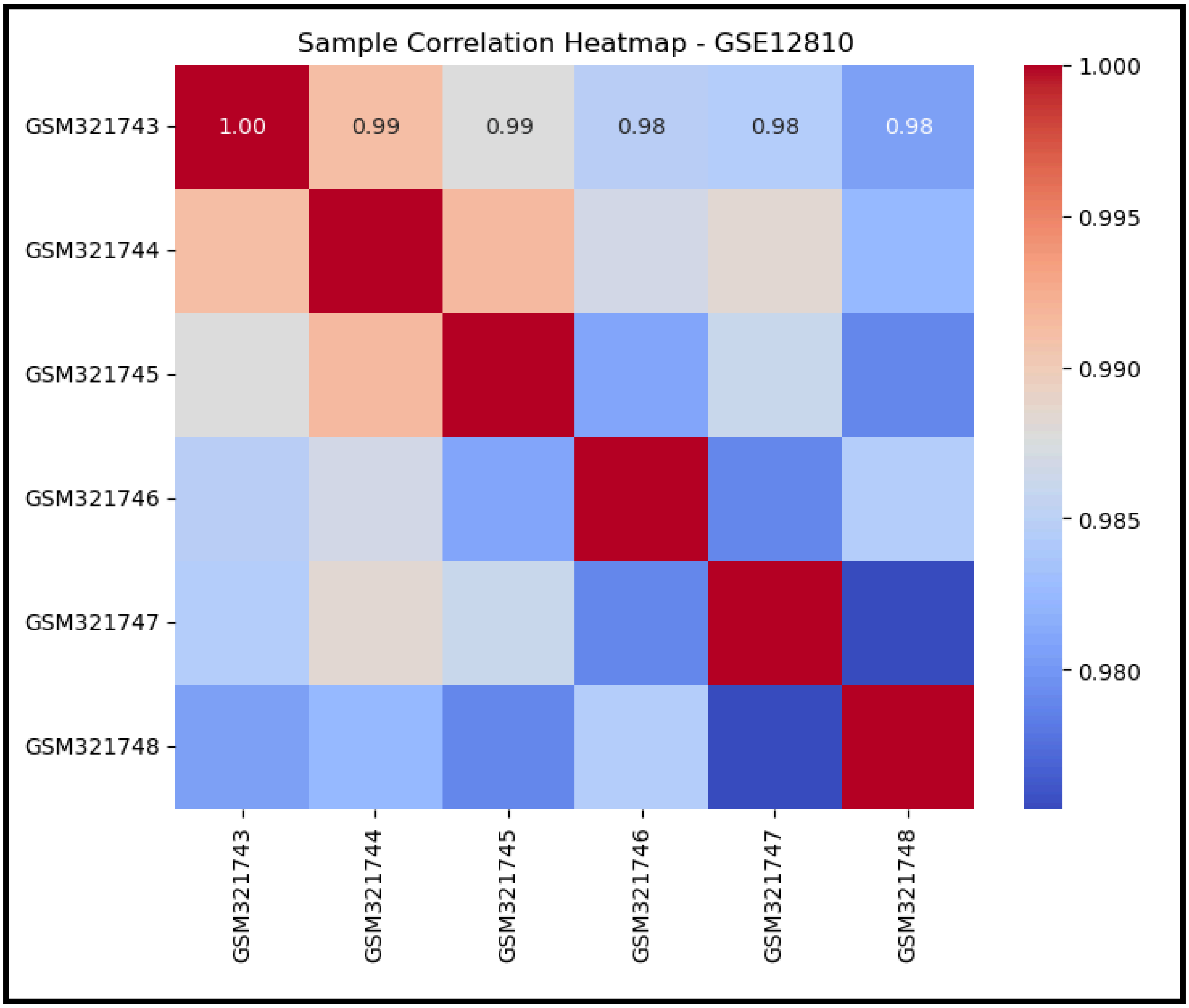


Figure 8. Sample correlation heatmap for GSE12810.

This heatmap shows high overall similarity between samples and supports the technical quality of the dataset.

A PCA plot was generated for GSE12810 to visualize control and noise-exposed samples. The PCA showed partial separation between noise-exposed and control samples, although the separation was not perfect. This suggested that GSE12810 contained a modest post-noise expression signal.

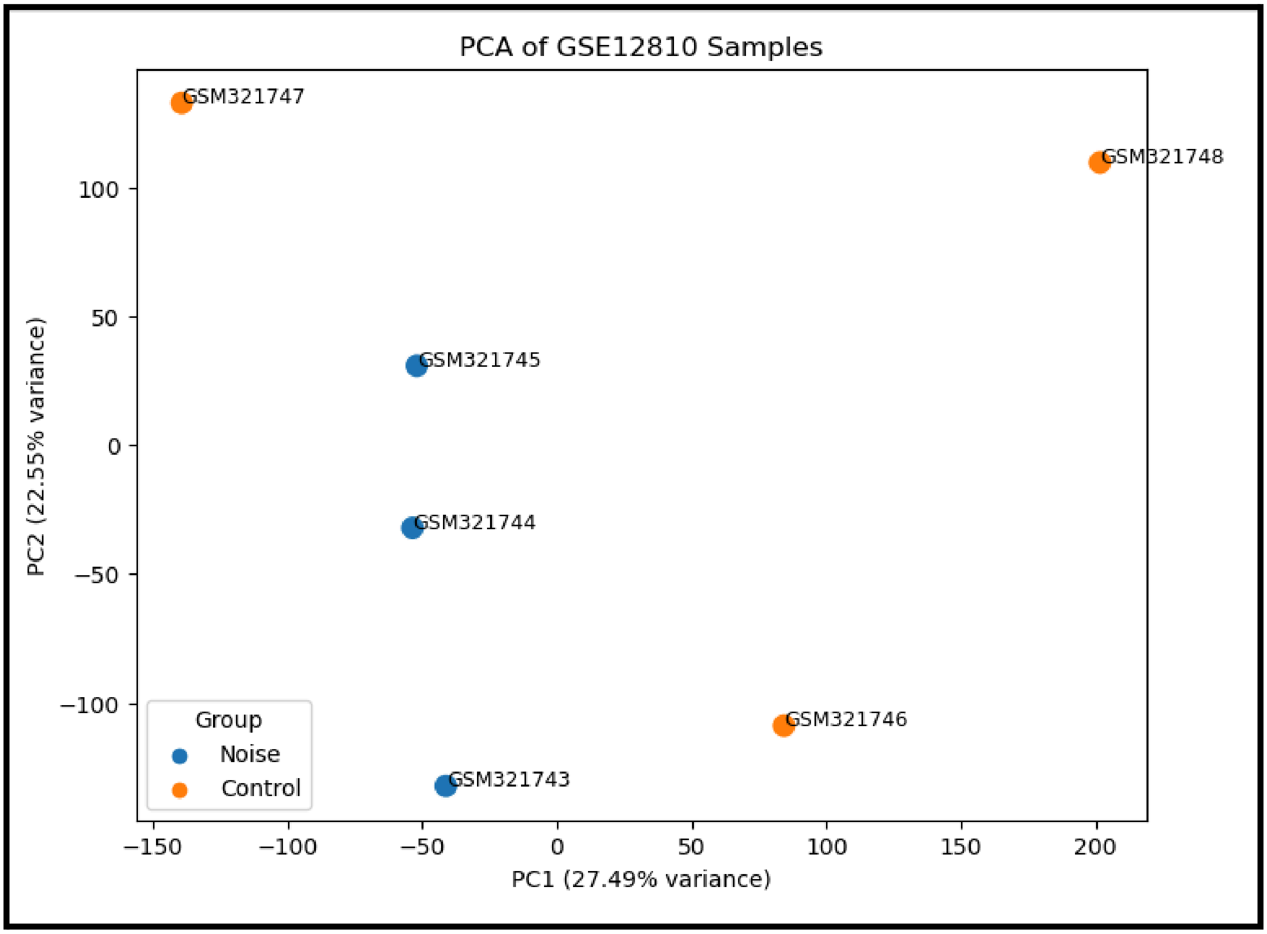


Figure 9. PCA plot of GSE12810 control and noise-exposed samples.

This plot shows partial separation between validation dataset groups.

Differential expression analysis was performed on GSE12810 using Welch's t-test and Benjamini-Hochberg false discovery rate correction. A total of 45,101 probes were analyzed. Under strict criteria using adjusted p-value and fold-change thresholds, 0 probes were significant. This result was likely influenced by the small sample size of 3 control and 3 noise-exposed samples.

Table 7. Strict differential expression summary for GSE12810.

Criterion	Result
Total probes analyzed	45,101
Strict significant probes after FDR correction	0
Main interpretation	Dataset underpowered for strict genome-wide discovery

Because no probes survived strict FDR correction, a relaxed ranked DEG table was created for validation-oriented interpretation. Using relaxed criteria of p-value < 0.05, absolute log2 fold change > 0.3, and annotated gene symbol present, 996 ranked expression-trend probes were identified. These were not treated as strict significant DEGs, but they helped identify expression trends relevant to validation. The GSE12810 validation analysis also identified 653 upregulated trend probes and 492 downregulated trend probes in the volcano trend output.

Table 8. GSE12810 relaxed ranked expression-trend genes.

Result Type	Count
Relaxed ranked DE probes	996
Upregulated trend probes	653
Downregulated trend probes	492

Strict FDR-significant probes	0
Interpretation	Useful for validation trends, not strict discovery

A GSE12810 volcano plot was generated using log2 fold change and p-value. Most probes clustered near zero, but both upregulated and downregulated trend probes were visible. This plot was used to visualize expression shifts in the validation dataset under relaxed trend thresholds.

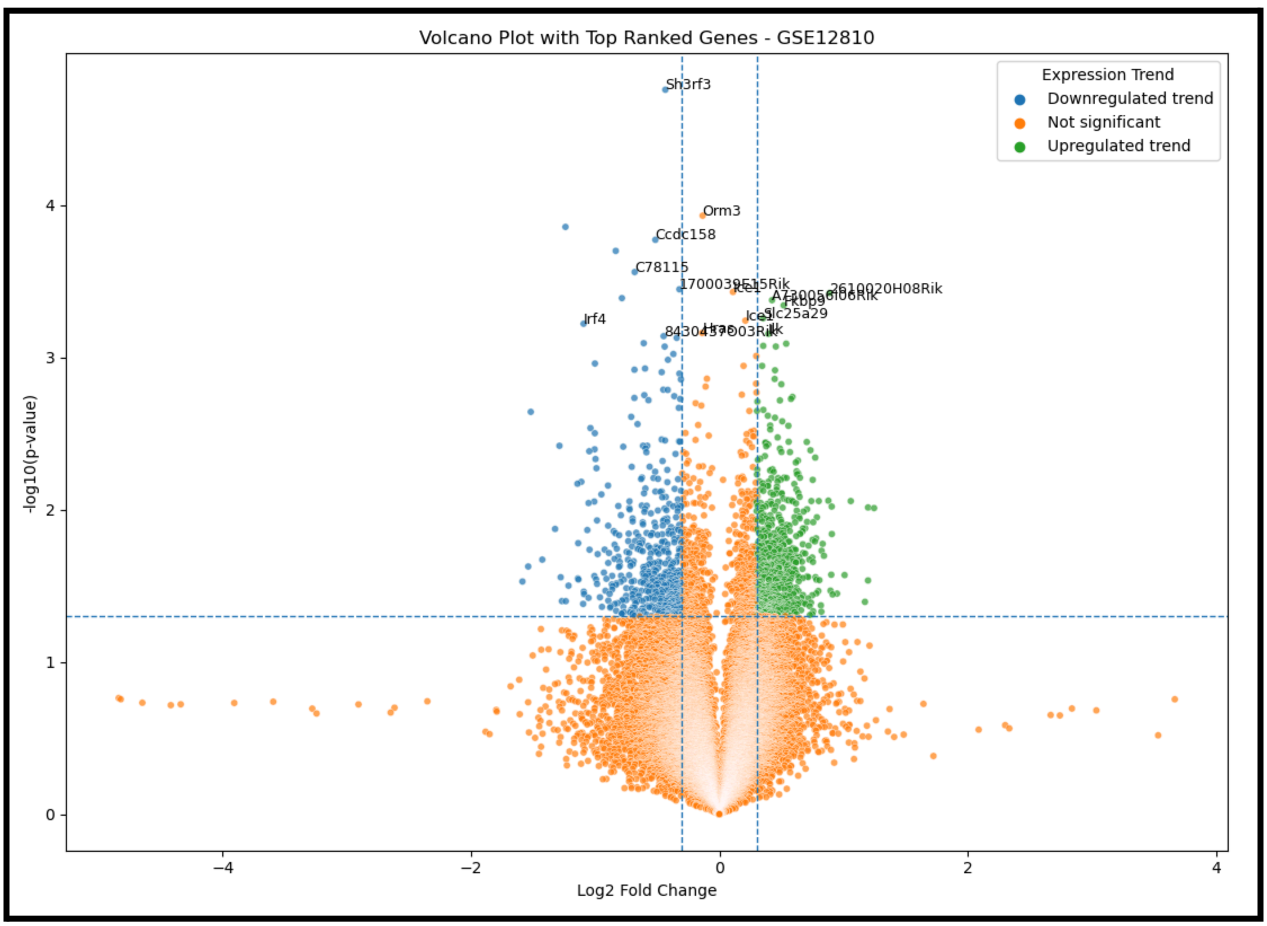
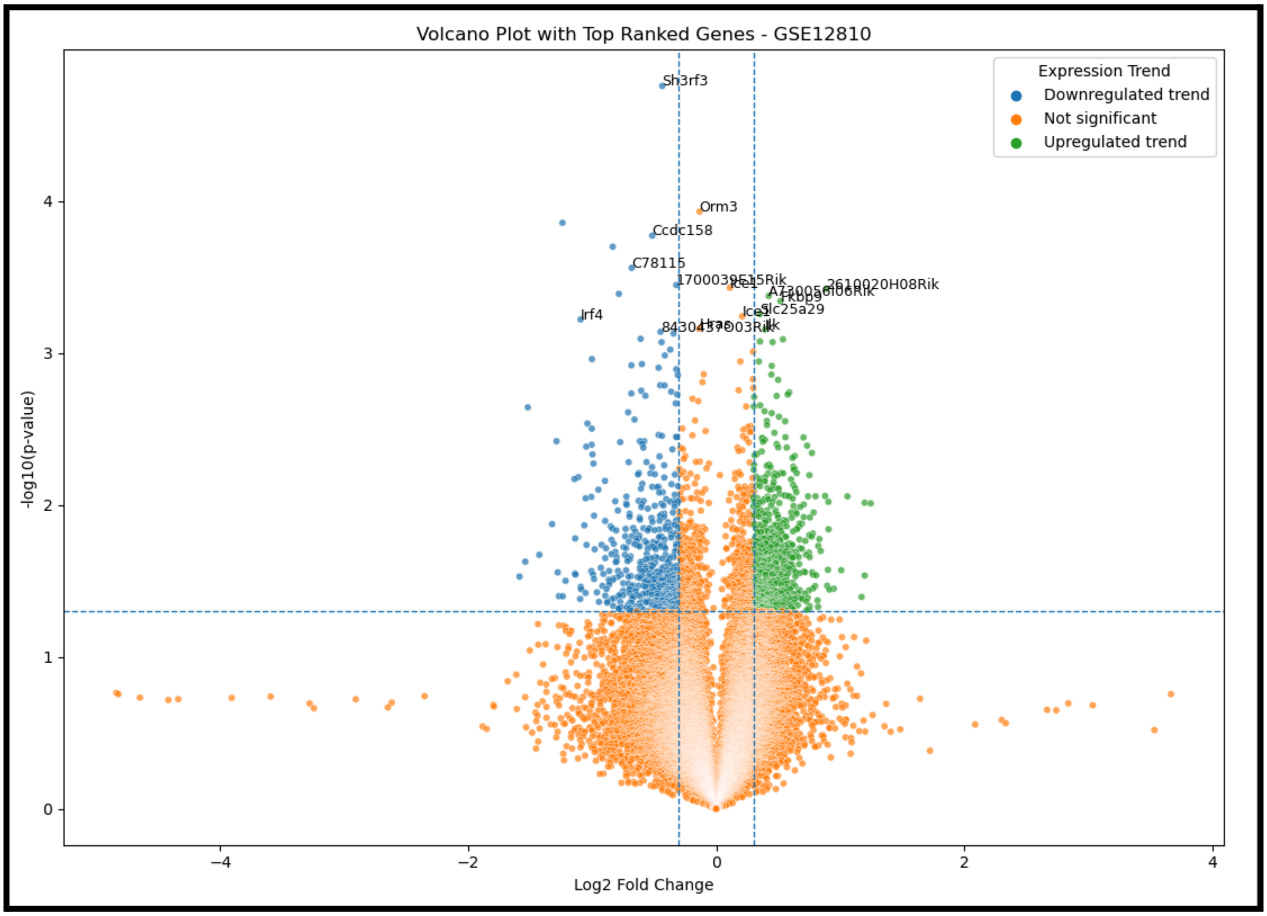


Figure 10. Volcano plot of GSE12810 differential expression trends.

This plot shows modest expression shifts in the validation dataset using relaxed trend thresholds.

A version of the GSE12810 volcano plot with top gene labels was also generated. This version labeled selected ranked genes to make the plot more interpretable.



Supplemental Figure S2. Labeled GSE12810 volcano plot.

This figure labels selected top-ranked trend genes from the validation dataset.

A top ranked DEG heatmap was also generated for GSE12810. This heatmap displayed expression trends among the highest-ranked genes from the relaxed differential expression analysis and was one of the clearest Dataset 2 visualizations.

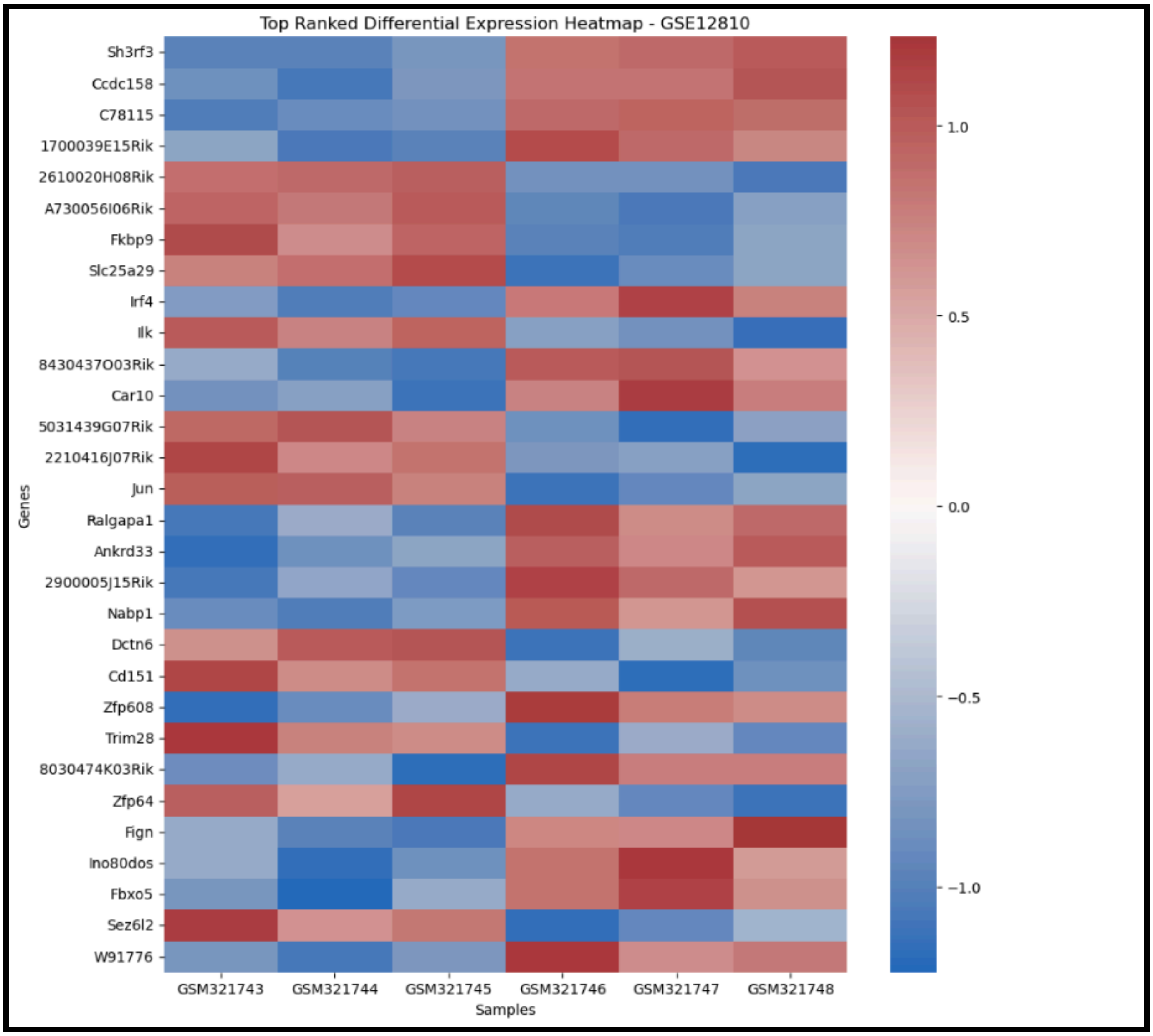


Figure 11. Heatmap of top ranked expression-trend genes in GSE12810.

This figure shows expression patterns among the highest-ranked validation dataset genes.

Dataset 1 candidate genes were then searched in GSE12810. The candidate genes tested were *Hspa1b, Dnm1l, Stat1, Gls, Dnaja2, Hnrnpu, Pde3b,* and *Srsf1*. All eight genes were found in GSE12810, with 46 total candidate probe hits. However, validation strength varied across genes.

Table 9. Best GSE12810 validation result for each Dataset 1 candidate gene.

Candidate	Best Match	Log2FC	p-value	Validation Label
Dnaja2	AK163592 /// *Dnaja2*	0.218887	0.285321	Weak/no validation trend
Dnm1l	*Dnm1l*	0.273567	0.220453	Weak/no validation trend
Gls	*Gls*	0.141196	0.090063	Weak/no validation trend
Hnrnpu	*Hnrnpul2*	0.063787	0.031246	Statistical trend only
Hspa1b	*Hspa1b*	0.476502	0.129344	Fold-change trend only
Pde3b	*Pde3b*	-0.097846	0.603912	Weak/no validation trend
Srsf1	LOC102641923 /// *Srsf1*	-0.175667	0.140910	Weak/no validation trend
Stat1	*Stat1*	-1.182340	0.067314	Fold-change trend only

The individual Dataset 1 candidates did not all show strong validation in GSE12810. However, partial support was observed for *Hnrnpu/Hnrnpul2, Hspa1b,* and *Stat1*. The stronger validation result came from the targeted oxidative stress/Nrf2 pathway-level analysis rather than from perfect one-to-one replication of every discovery candidate.

A curated oxidative stress/Nrf2-related gene set was tested in GSE12810. This set included genes involved in core Nrf2 signaling, antioxidant defense, ROS detoxification, mitochondrial stress, apoptosis, inflammatory response, and heat-shock response. This analysis produced 280 oxidative stress/Nrf2-related probe hits across 48 unique target genes. The validation categories included 6 strong validation trend genes, 10 fold-change trend genes, 1 statistical trend gene, and 31 weak/no validation trend genes.

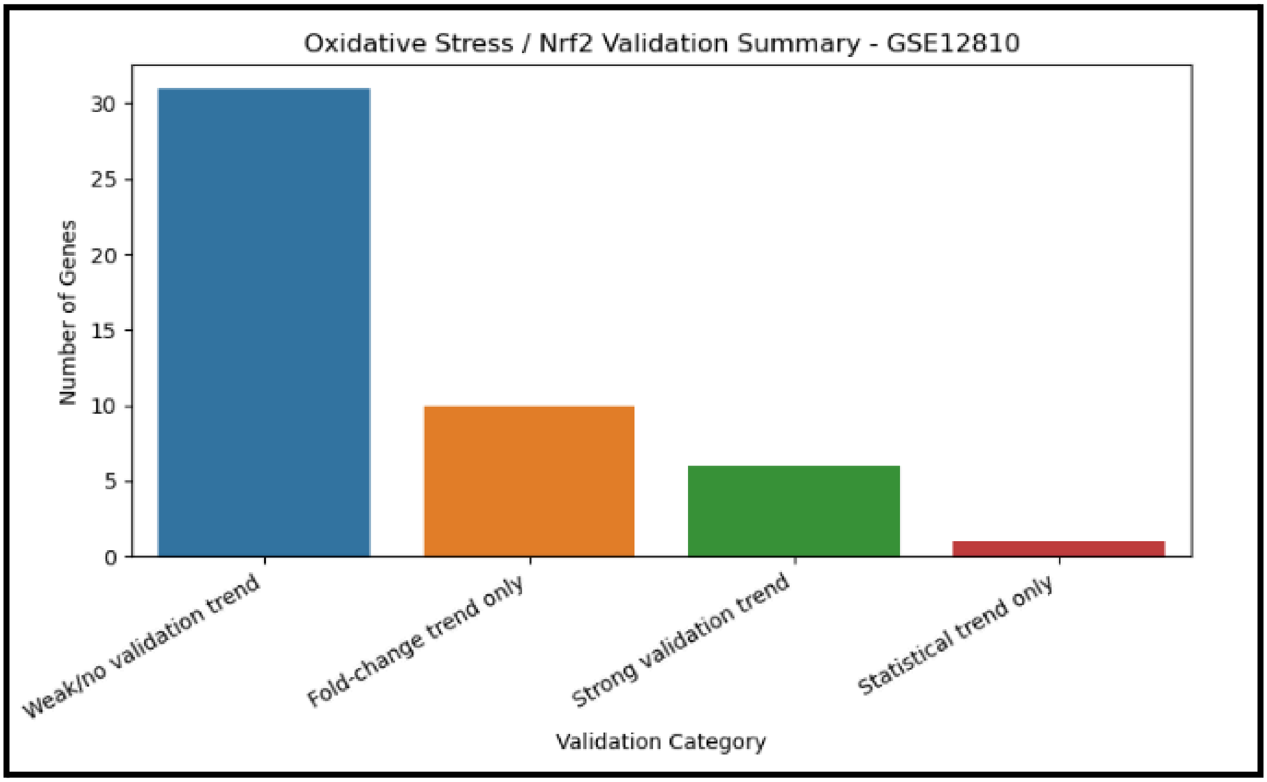


Figure 12. GSE12810 oxidative stress/Nrf2 validation category summary.

This bar plot summarizes the validation categories for oxidative stress and Nrf2-related genes.

Table 10. GSE12810 oxidative stress/Nrf2 validation summary.

Validation Category	**Number of Genes**
Strong validation trend	6
Fold-change trend only	10
Statistical trend only	1

Weak/no validation trend	31
Total target genes found	48

The strong validation trend genes in GSE12810 included *Irf4, Jun, Tnfaip2, Bcl2l12, Il6st,* and *Hspb11*. Important partial validation genes included *Sod1, Stat1, Hspa1b, Gpx1, Nfe2l2,* and *Casp3*.

Table 11. Selected oxidative stress/Nrf2-related validation genes in GSE12810.

Gene	**Result**
Irf4	Strong validation trend
Jun	Strong validation trend
Tnfaip2	Strong validation trend
Bcl2l12	Strong validation trend
Il6st	Strong validation trend
Hspb11	Strong validation trend
Sod1	Statistical trend
Stat1	Fold-change trend
Hspa1b	Fold-change trend

Gpx1	Fold-change trend
Nfe2l2	Fold-change trend
Casp3	Fold-change trend

An oxidative stress/Nrf2 heatmap was also generated for GSE12810. This heatmap showed expression patterns for genes related to oxidative stress, Nrf2 signaling, mitochondrial stress, apoptosis, inflammatory signaling, and heat-shock response. The separation between control and noise-exposed samples was not perfect, but several stress-related genes showed visible expression differences.

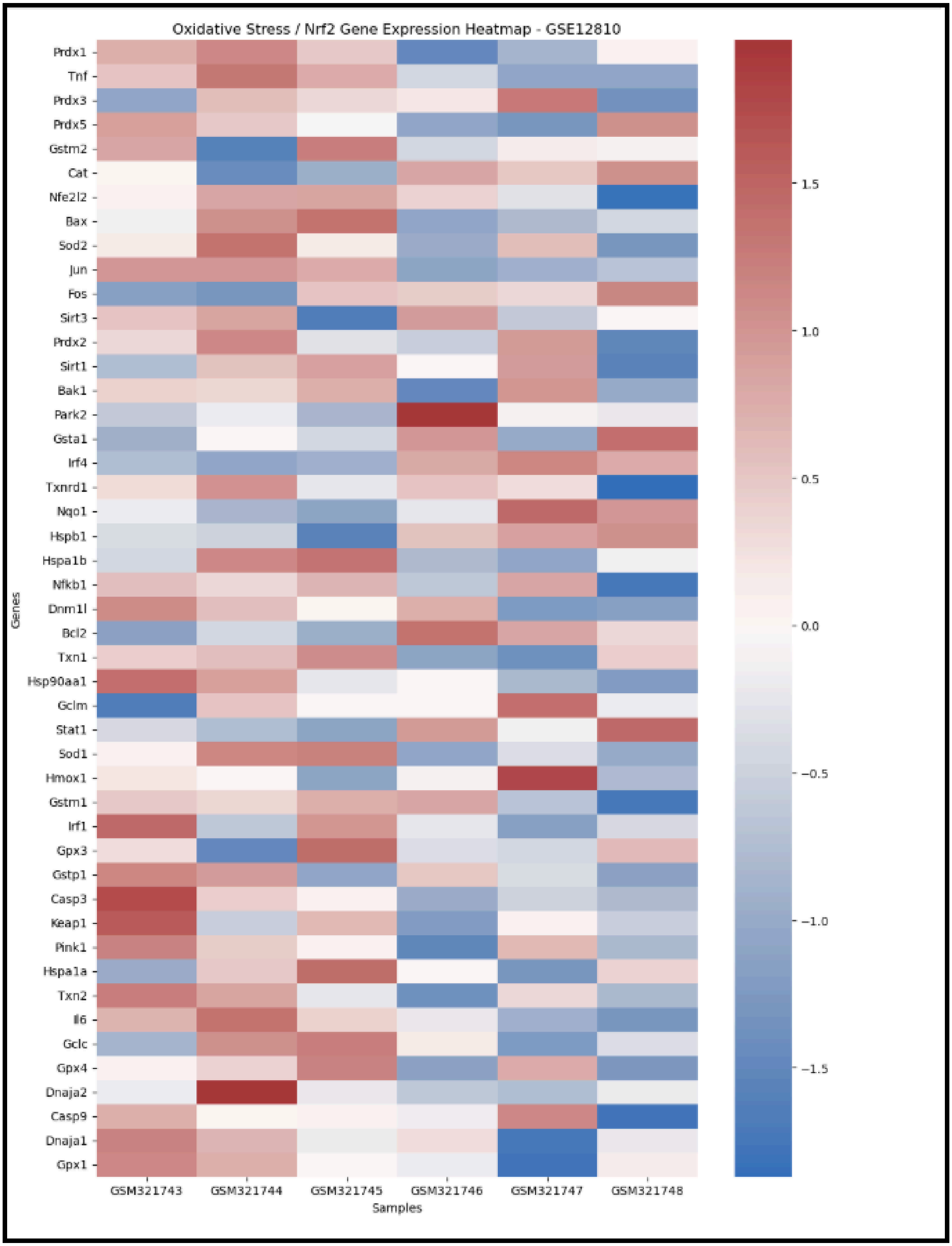

Oxidative Stress / Nrf2 Gene Expression Heatmap - GSE12810
Genes
Prdx1
Tnf
Prdx3
Prdx5
Gstm2
Cat
Nfe2l2
Bax
Sod2
Jun
Fos
Sirt3
Prdx2
Sirt1
Bak1
Park2
Gsta1
Irf4
Txnrd1
Nqo1
Hspb1
Hspa1b
Nfkb1
Dnm1l
Bcl2
Txn1
Hsp90aa1
Gclm
Stat1
Sod1
Hmox1
Gstm1
Irf1
Gpx3
Gstp1
Casp3
Keap1
Pink1
Hspa1a
Txn2
Il6
Gclc
Gpx4
Dnaja2
Casp9
Dnaja1
Gpx1
GSM321743
GSM321744
GSM321745
GSM321746
GSM321747
GSM321748
Samples
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5

Figure 13. Oxidative stress/Nrf2 gene heatmap for GSE12810.

This heatmap shows expression patterns among targeted oxidative stress and Nrf2-related genes in the validation dataset.

Cross-Dataset Validation Results

After completing the discovery and validation analyses separately, GSE100365 and GSE12810 were compared directly. Each dataset was collapsed to the best probe per gene based on p-value, and shared genes were merged by gene symbol.

Across the whole transcriptome, 20,226 overlapping genes were identified between GSE100365 and GSE12810. Of these, 10,116 genes showed the same direction of expression change, while 10,110 genes showed the opposite direction. This produced approximately 50.01% direction agreement, indicating that the two datasets did not strongly replicate each other across the entire transcriptome.

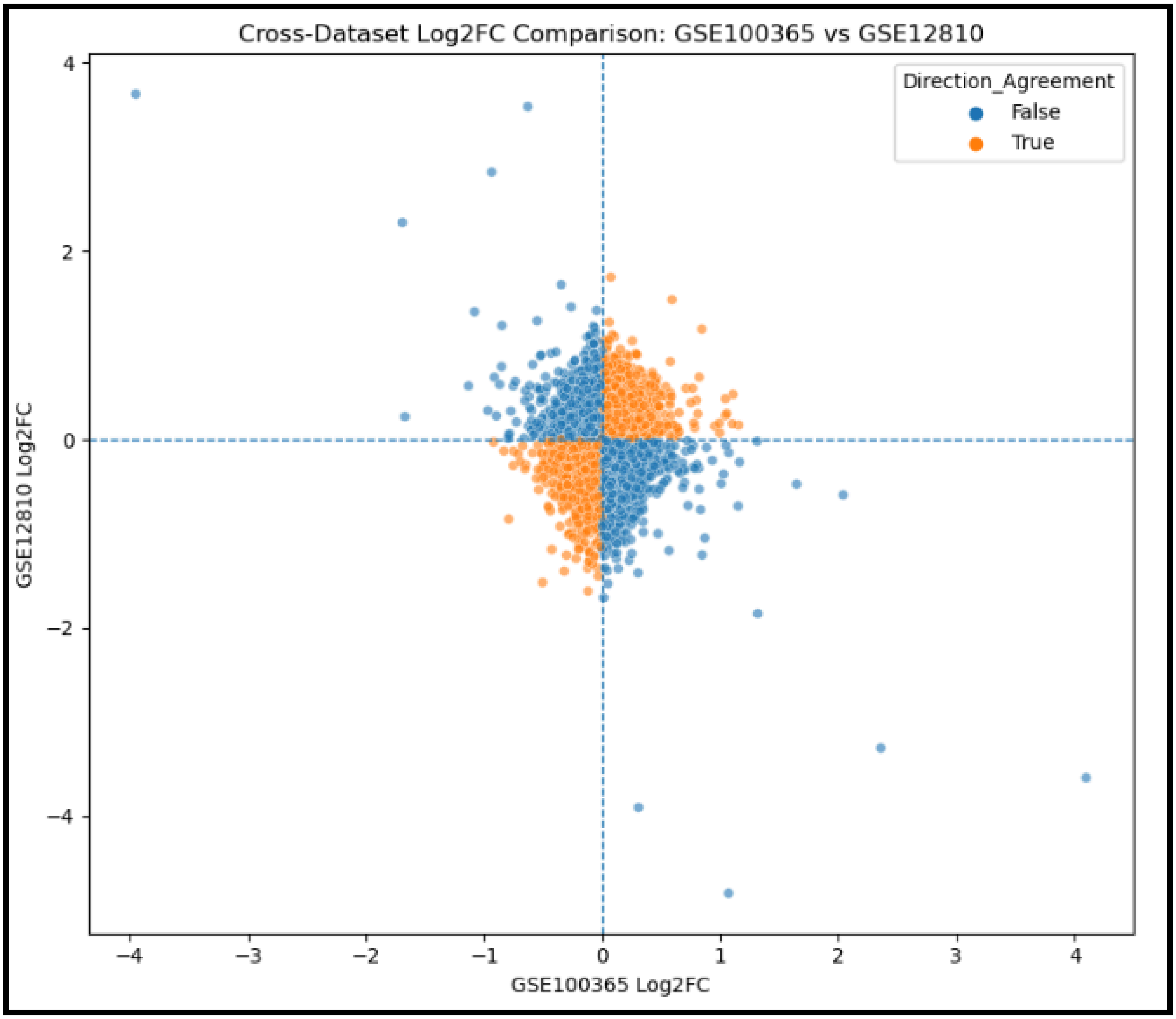


Figure 14. Whole-transcriptome cross-dataset Log2FC comparison between GSE100365 and GSE12810.

This scatterplot compares gene-level log2 fold-change values between the discovery and validation datasets.

Table 12. Whole-transcriptome cross-dataset direction agreement.

Comparison	**Count**

Overlapping genes	20,226
Same direction	10,116
Opposite direction	10,110
Percent direction agreement	50.01%

Because the two datasets differed in biological context, tissue focus, and experimental timing, a more targeted cross-dataset comparison was performed using oxidative stress/Nrf2-related genes. This targeted comparison identified 46 overlapping oxidative stress/Nrf2-related genes between datasets. Of these, 25 showed the same direction of expression change and 21 showed opposite directions, producing 54.35% direction agreement. This represented a modest improvement over the whole-transcriptome comparison.

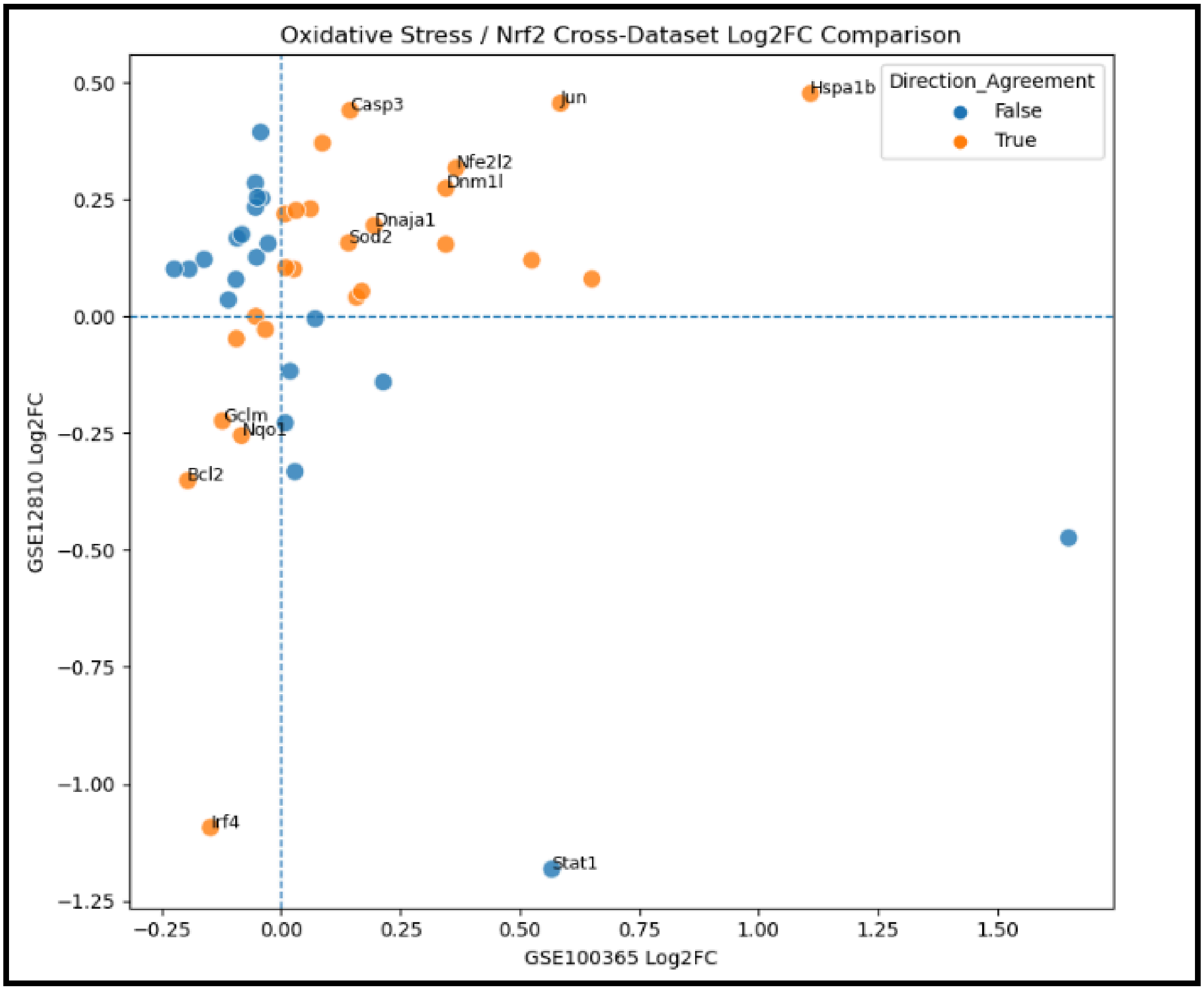


Figure 15. Oxidative stress/Nrf2 cross-dataset Log2FC comparison.

This scatterplot compares log2 fold-change values for oxidative stress and Nrf2-related genes across GSE100365 and GSE12810.

Table 13. Oxidative stress/Nrf2 cross-dataset direction agreement.

Comparison	Count

Overlapping oxidative/Nrf2 genes	46
Same direction	25
Opposite direction	21
Percent direction agreement	54.35%

A ranked oxidative/Nrf2 cross-dataset validation table was generated using a combined validation score based on fold-change magnitude in both datasets and agreement in expression direction. This ranking helped identify the strongest cross-dataset oxidative stress/Nrf2-related genes.

Table 14. Ranked oxidative stress/Nrf2 cross-dataset validation genes.

Final Validated Gene	Biological Relevance
Hspa1b	**Heat-shock response and cellular stress adaptation**
Jun	**Stress-response transcription signaling**
Nfe2l2	**Nrf2 antioxidant pathway regulation**
Dnm1l	**Mitochondrial fission and mitochondrial stress**
Casp3	**Apoptosis execution**

Bcl2	**Cell survival and apoptosis regulation**
Dnaja1	**Chaperone response and protein-folding stress**
Gclm	**Glutathione synthesis and antioxidant defense**
Nqo1	**Nrf2-related detoxification and antioxidant response**
Sod2	**Mitochondrial antioxidant defense**
Pink1	**Mitochondrial quality control and mitophagy**
Sirt3	**Mitochondrial redox regulation**

This table should include the top-ranked genes by final validation score, including *Hspa1b, Jun, Nfe2l2, Dnm1l, Casp3, Bcl2, Dnaja1, Gclm, Nqo1, Sod2, Pink1,* and *Sirt3*.

A final validated oxidative stress/Nrf2 gene heatmap was generated using the strongest cross-dataset validation genes. These genes were selected based on directional agreement, biological relevance, and connection to oxidative stress, mitochondrial dysfunction, apoptosis, heat-shock response, antioxidant defense, and Nrf2 signaling.

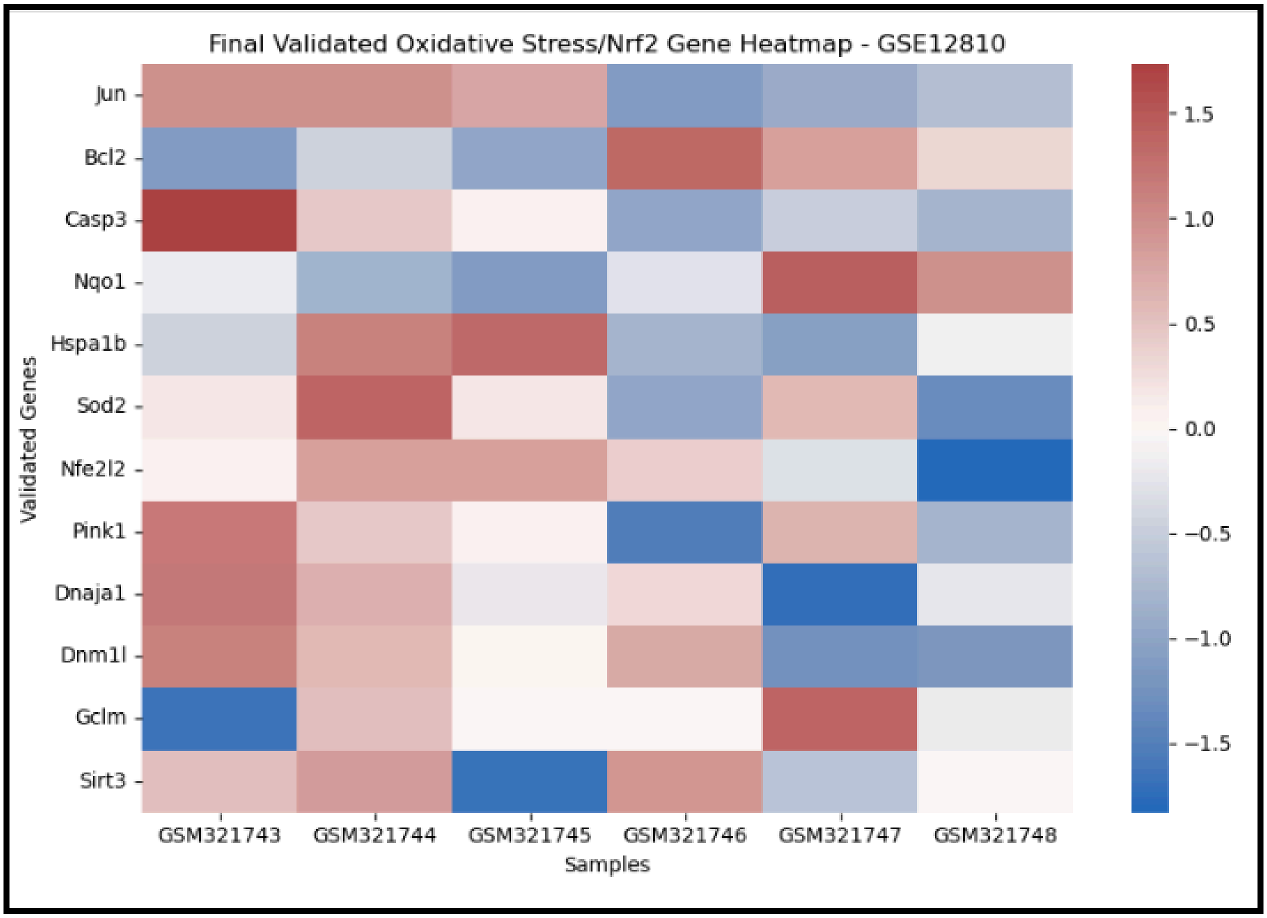


Figure 16. Final validated oxidative stress/Nrf2 gene heatmap.

This heatmap shows expression patterns for the final validated gene set in the validation dataset.

A final log2 fold-change comparison bar plot was generated to compare the final validated genes across GSE100365 and GSE12810. This plot showed whether each final gene was upregulated or downregulated in each dataset.

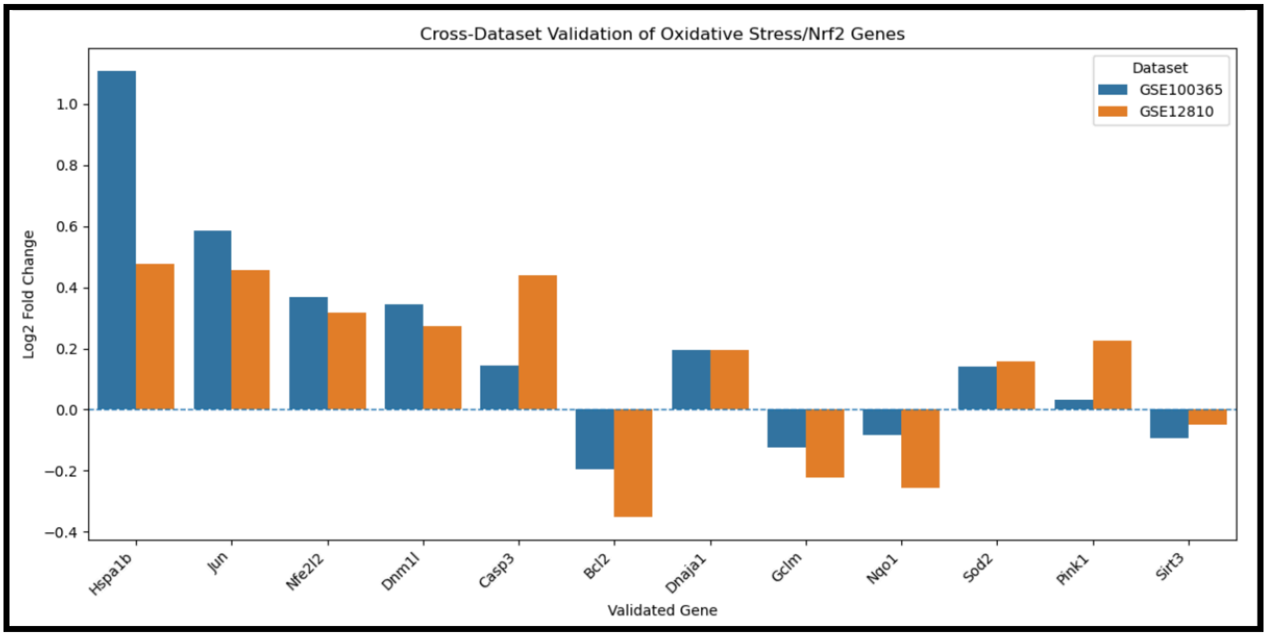


Figure 17. Final validated oxidative stress/Nrf2 gene log2 fold-change comparison.

This plot compares the direction and magnitude of expression change for the final validated gene set across both datasets.

Final Validated Oxidative Stress/Nrf2 Gene Set

The final validated oxidative stress/Nrf2-related gene set consisted of 12 genes:

Hspa1b, Jun, Nfe2l2, Dnm1l, Casp3, Bcl2, Dnaja1, Gclm, Nqo1, Sod2, Pink1, and *Sirt3*.

These genes were selected based on cross-dataset directional agreement, biological relevance, oxidative stress/Nrf2 pathway connection, mitochondrial stress relevance, apoptosis relevance, heat-shock response relevance, and redox balance relevance. The final validated gene set included genes involved in heat-shock response, stress-response transcriptional regulation, Nrf2-mediated antioxidant signaling, mitochondrial dysfunction, apoptosis regulation, mitochondrial quality control, and redox homeostasis.

Table 15. Final validated oxidative stress/Nrf2-related gene set.

Gene	Biological Category	Main Biological Interpretation	Direction Pattern
Hspa1b	Heat-shock response	Cellular stress adaptation and proteostasis	Up in both datasets
Jun	Stress-response transcription	Injury signaling and transcriptional stress response	Up in both datasets
Nfe2l2	Nrf2 regulation	Antioxidant-response regulator	Up in both datasets
Dnm1l	Mitochondrial dynamics	Mitochondrial fission and ROS-linked dysfunction	Up in both datasets
Casp3	Apoptosis	Execution of apoptosis pathway	Up in both datasets
Bcl2	Apoptosis regulation	Cell survival and apoptosis balance	Down in both datasets
Dnaja1	Chaperone response	Protein-folding stress and cellular protection	Up in both datasets
Gclm	Antioxidant defense	Glutathione synthesis and redox balance	Down in both datasets

Nqo1	Nrf2 antioxidant response	Detoxification and antioxidant enzyme activity	Down in both datasets
Sod2	Mitochondrial antioxidant defense	Mitochondrial superoxide detoxification	Up in both datasets
Pink1	Mitochondrial quality control	Mitophagy and mitochondrial stress response	Up in both datasets
Sirt3	Mitochondrial redox regulation	Mitochondrial stress resistance and redox control	Down in both datasets

Overall, the data showed that GSE100365 provided the strongest discovery signal, while GSE12810 provided partial but meaningful pathway-level validation. The whole-transcriptome comparison did not show strong global replication, but the oxidative stress/Nrf2-focused comparison showed modestly stronger agreement and supported the final focus on oxidative stress, mitochondrial dysfunction, apoptosis regulation, heat-shock response, and Nrf2-related antioxidant defense.

Supplemental Figures and Tables

Additional supplemental figures and tables were generated during the analysis but were not all included in the final paper due to space and readability. The main Data section includes the strongest figures and tables needed to support the results.

ANALYSIS

The results of this project support the hypothesis that noise exposure is associated with altered expression of oxidative stress and Nrf2-related genes. However, the results also show that NIHL is not explained by one isolated gene or a simple one-pathway model. Instead, the strongest interpretation is that noise exposure produces a coordinated biological response involving oxidative stress, mitochondrial dysfunction, apoptosis regulation, heat-shock response, inflammatory signaling, RNA-processing changes, and antioxidant defense. This systems-level pattern fits the background research, which describes NIHL as a multi-layered condition involving reactive oxygen species, mitochondrial injury, apoptotic signaling, antioxidant pathway activation, and cochlear stress responses.

The discovery dataset, GSE100365, provided the strongest evidence for broad transcriptomic disruption after noise exposure. The PCA results showed that control and noise-exposed samples partially separated, meaning that their overall gene expression profiles were different. The timepoint PCA was especially important because it showed that noise-exposed samples did not form one uniform cluster. Instead, samples from different post-noise timepoints separated from each other, suggesting that the biological response to acoustic trauma changes over time. This matters because NIHL is not a single moment of damage. It may involve immediate injury, delayed oxidative stress, repair attempts, inflammation, and later degeneration. Therefore, collapsing all noise-exposed samples into one group made the main comparison easier, but it also simplified a time-dependent biological process.

The heatmap and volcano plot from GSE100365 supported the same overall conclusion. The z-score heatmap showed that the top differentially expressed genes had different expression patterns across control and noise-exposed samples. The volcano plot helped identify genes with both statistical and biological relevance based on p-value and approximate log2 fold change. Together, these figures showed that noise exposure was associated with measurable transcriptomic changes, especially in genes related to mitochondrial regulation, cellular stress response, RNA processing, and inflammatory signaling. This supports the idea that noise-induced injury affects several connected cellular systems rather than only classical antioxidant genes.

Several important genes from Dataset 1 helped shape the biological interpretation. *Dnm1l* was one of the strongest mechanistic candidates because it is involved in mitochondrial fission. Since mitochondria are both a source and target of reactive oxygen species, altered *Dnm1l* expression suggests that mitochondrial dynamics may be part of the cochlear injury response. *Hspa1b* and *Dnaja2* pointed toward heat-shock response and protein-folding stress, suggesting that cells may activate protective chaperone systems after noise exposure. *Stat1* suggested inflammatory or immune-related signaling, while *Hnrnpu* and *Srsf1* suggested changes in RNA processing and post-transcriptional regulation. These results expand the interpretation beyond a simple "oxidative stress gene list" and show that NIHL may involve a larger stress-response network. The master project summary identifies these same themes as key Dataset 1 findings, including mitochondrial regulation, heat-shock response, RNA regulation, inflammatory signaling, and cellular stress response.

The Random Forest analysis was useful, but it also revealed one of the most important statistical limitations of the project. At first, the model appeared to have an accuracy of approximately 0.80, which could suggest that gene expression profiles were able to classify control and noise-exposed samples. However, deeper evaluation showed that the model mainly predicted the majority class, meaning it classified most or all samples as noise-exposed. Because GSE100365 had 12 noise-exposed samples but only 3 control samples, the model's apparent accuracy was inflated by class imbalance. This means the Random Forest model should not be interpreted as a reliable diagnostic classifier. Instead, it is more appropriate to interpret it as an exploratory feature-prioritization tool. Its feature-importance results were still useful because several important genes, including *Hspa1b* and *Stat1*, were biologically connected to stress response and inflammation.

The Gene Impact Score helped address one weakness of relying only on p-values. In high-dimensional gene expression data, a gene can have a low p-value but limited biological relevance, or it can have moderate statistical evidence but strong biological importance. The Gene Impact Score combined multiple forms of evidence, including differential expression, fold-change magnitude, Random Forest feature importance, biological relevance, and network centrality. This made the ranking more biologically meaningful. Gene Impact Score v2 was stronger than v1 because it included network centrality, allowing genes to be prioritized not only by individual expression change but also by their position within the co-expression network. The fact that *Hnrnpu, Hspa1b, Dnm1l, Stat1, Dnaja2, Pde3b,* and *Srsf1* remained important across multiple layers of analysis supports the idea that NIHL involves coordinated gene networks rather than independent gene changes.

The co-expression network also supported a systems-level interpretation. The network contained 22 nodes and 48 edges, meaning that many top candidate genes had correlated expression patterns. Although this network was exploratory because the sample size was small, it suggested that the genes identified in the project may act in connected modules. This fits the biology of NIHL because oxidative stress, mitochondrial dysfunction, apoptosis, inflammation, and heat-shock response are not separate processes. They influence each other. For example, mitochondrial dysfunction can increase ROS production, ROS can activate apoptotic signaling, apoptotic stress can trigger inflammatory responses, and antioxidant pathways such as Nrf2 can respond to this damage. The network analysis therefore gave the project a more integrated interpretation than a simple ranked gene table would have.

The validation dataset, GSE12810, produced a more cautious but still meaningful result. Under strict genome-wide criteria using adjusted p-value and fold-change thresholds, GSE12810 had 0 significant probes. This result does not mean that the validation dataset was useless or that the hypothesis failed. Instead, it reflects the difficulty of detecting statistically significant gene expression changes in a very small dataset with only 3 noise-exposed and 3 control samples. Since 45,101 probes were tested, false discovery rate correction was extremely strict. As a result, GSE12810 was underpowered for full genome-wide discovery. It was more useful for ranked validation, candidate-gene checking, and pathway-level interpretation than for identifying strict significant DEGs.

The GSE12810 PCA and heatmaps still suggested that the validation dataset contained a biological signal. The PCA showed partial separation between control and noise-exposed samples, and the sample correlation heatmap showed that the samples were highly correlated without major outliers. This means the dataset appeared technically clean, even if it was statistically underpowered. The relaxed ranked DEG analysis identified 996 ranked expression-trend probes, which were not treated as strict significant genes but were useful for validation-oriented interpretation. This distinction is important because it prevents overclaiming. The validation dataset did not prove every discovery gene, but it did help test whether oxidative stress and Nrf2-related patterns appeared in an independent noise-related dataset.

The individual Dataset 1 candidate validation results were mixed. All eight discovery candidate genes tested in GSE12810 were found in the validation dataset, producing 46 candidate probe hits. However, none of the eight showed a strong validation trend under the labeling system used. *Hnrnpu/Hnrnpul2* showed a statistical trend only, while *Hspa1b* and *Stat1* showed fold-change trends only. The other candidates showed weak or no validation trend. This means the validation did not support a perfect one-to-one replication of every discovery gene. However, that result is not surprising because the datasets differed in sample size, tissue context, timing, and experimental design. The stronger validation story came from the pathway-level oxidative stress/Nrf2 analysis rather than individual-gene replication.

The targeted oxidative stress/Nrf2 validation results were more meaningful than the whole-transcriptome comparison. In GSE12810, the oxidative stress/Nrf2 candidate analysis found 280 probe hits across 48 unique target genes. Of those 48 genes, 6 showed strong validation trends, 10 showed fold-change trends, 1 showed a statistical trend only, and 31 showed weak or no validation trend. This means 17 of 48 oxidative stress/Nrf2-related genes showed at least some form of validation support. The strong validation trend genes included *Irf4, Jun, Tnfaip2, Bcl2l12, Il6st,* and *Hspb11*, while important partial validation genes included *Sod1, Stat1, Hspa1b, Gpx1, Nfe2l2,* and *Casp3*. These results support the idea that the validation dataset was more informative at the pathway level than at the level of individual discovery genes. The GSE12810 validation analysis specifically shows 45,101 probes analyzed, 0 strict FDR-significant probes, 996 relaxed ranked DE probes, 46 Dataset 1 candidate probe hits, and targeted oxidative stress/Nrf2 validation across 48 genes.

The cross-dataset comparison also supported a pathway-level interpretation. Across the whole transcriptome, 20,226 overlapping genes were identified between GSE100365 and GSE12810. Of these, 10,116 genes changed in the same direction, while 10,110 changed in the opposite direction. This produced about 50.01% direction agreement, which is essentially close to an even split. In other words, the datasets did not strongly replicate each other across the entire transcriptome. However, when the comparison was narrowed to oxidative stress/Nrf2-related genes, 46 overlapping genes were identified, with 25 showing the same direction and 21 showing the opposite direction. This produced 54.35% direction agreement. Although this is still modest, it is stronger than the whole-transcriptome comparison and suggests that the targeted pathway had more consistent cross-dataset behavior than the overall gene expression background.

The final validated oxidative stress/Nrf2 gene set was selected from this combined evidence rather than from one single statistical test. The final set consisted of *Hspa1b, Jun, Nfe2l2, Dnm1l,* *Casp3, Bcl2, Dnaja1, Gclm, Nqo1, Sod2, Pink1,* and *Sirt3*. These genes represent several major biological categories. *Hspa1b* and *Dnaja1* represent heat-shock and chaperone response. *Jun* represents stress-response transcriptional signaling. *Nfe2l2, Gclm,* and *Nqo1* connect directly to Nrf2-mediated antioxidant regulation. *Dnm1l, Sod2, Pink1,* and *Sirt3* connect to mitochondrial dynamics, mitochondrial antioxidant defense, mitophagy, and redox regulation. *Casp3* and *Bcl2* represent apoptosis and cell survival balance. This final gene set is important because it does not reduce NIHL to one marker. Instead, it reflects a coordinated injury-response signature involving mitochondrial stress, oxidative defense, apoptosis regulation, and cellular stress adaptation.

The direction patterns in the final gene set also suggest biological complexity. Several genes, including *Hspa1b, Jun, Nfe2l2, Dnm1l, Casp3, Dnaja1, Sod2,* and *Pink1*, were interpreted as upregulated in both datasets. This suggests activation of stress response, mitochondrial remodeling, antioxidant regulation, apoptosis, and mitochondrial quality control after noise exposure. Other genes, including *Bcl2, Gclm, Nqo1,* and *Sirt3*, were interpreted as downregulated in both datasets. These downregulated patterns may reflect weakened protective antioxidant or survival responses, although this interpretation should be cautious because microarray datasets cannot prove causation. Still, the mixture of upregulated injury-response genes and downregulated protective genes fits a model in which cells are responding to damage but may not be fully able to restore redox balance.

Overall, the results partially support the original hypothesis. The project did find altered expression patterns in genes related to oxidative stress, mitochondrial function, apoptosis, heat-shock response, and Nrf2-related antioxidant defense. However, the results were not uniformly strong across every dataset, every gene, or every analysis method. GSE100365 provided strong discovery evidence, while GSE12810 provided partial but meaningful validation. The strongest conclusion is not that every oxidative stress gene was significantly changed, but that oxidative stress and Nrf2-related pathways appeared repeatedly across multiple layers of analysis.

Several limitations must be considered. First, both datasets used mouse tissue, so the results may not fully translate to human NIHL. Second, both datasets had small sample sizes, especially GSE12810 with only 6 samples total. Third, GSE100365 had class imbalance, with 12 noise-exposed samples and only 3 controls, which affected machine learning performance. Fourth, the noise-exposed samples in GSE100365 came from multiple timepoints, and collapsing them into one noise group may have hidden time-specific gene expression patterns. Fifth, GSE12810 did not produce strict FDR-significant genes, so its results had to be interpreted through relaxed trends and pathway-level validation. Sixth, this project analyzed gene expression rather than protein levels, enzyme activity, or direct oxidative stress measurements. Therefore, the results show transcriptomic associations, not direct proof of biological mechanism.

Another important limitation is that genetic variation was not directly analyzed. The original research question included genetic variation because individual susceptibility to NIHL may be influenced by variants in antioxidant defense genes, Nrf2 pathway genes, mitochondrial genes, or apoptosis-related genes. However, this project did not perform SNP analysis, GWAS analysis, variant annotation, or genotype-expression integration. Because of this, genetic variation should not be presented as a completed result. Instead, it should be treated as a future direction. A future version of this project could connect the final 12-gene oxidative stress/Nrf2 signature to known genetic variants, expression quantitative trait loci, or human NIHL susceptibility studies.

Future research could improve this project in several ways. A larger dataset with more balanced control and noise-exposed groups would make the differential expression and machine learning results more reliable. A timepoint-specific analysis of GSE100365 could help identify which genes are involved in immediate injury, delayed oxidative stress, recovery, or long-term degeneration. RNA-seq datasets could provide higher resolution than microarray data. Protein-level validation could test whether transcriptomic changes correspond to actual changes in antioxidant enzymes, heat-shock proteins, apoptotic markers, or mitochondrial proteins. Experimental validation using qPCR, Western blotting, or functional assays would be needed to confirm the biological role of the final gene set. Finally, integrating genetic variation data would help answer the original susceptibility question more directly.

In conclusion, the analysis shows that NIHL is best understood as a coordinated molecular stress response rather than a single-gene condition. The final validated gene set reflects oxidative stress, mitochondrial dysfunction, Nrf2-related antioxidant signaling, apoptosis regulation, and heat-shock response. The results support the project's central hypothesis while also showing the limits of small public datasets and exploratory machine learning. This project provides a strong computational foundation for future studies that could experimentally validate these genes and connect transcriptomic patterns to genetic susceptibility in NIHL.

CONCLUSION

The hypothesis for this study was supported. Noise-exposed auditory tissue showed altered gene expression patterns compared with control tissue, and many of the most meaningful changes were connected to oxidative stress, mitochondrial dysfunction, apoptosis regulation, heat-shock response, inflammatory signaling, and Nrf2-related antioxidant defense. Although the results did not show that every individual oxidative stress gene changed significantly in both datasets, the overall pathway-level pattern supported the idea that NIHL is associated with coordinated molecular stress responses rather than one isolated gene.

GSE100365 served as the strongest discovery dataset. It showed broad transcriptomic differences between control and noise-exposed samples through differential expression analysis, PCA, heatmap visualization, volcano plot analysis, Random Forest feature prioritization, Gene Impact Score ranking, and co-expression network analysis. Several important genes identified in the discovery phase, including *Hspa1b, Dnm1l, Stat1, Dnaja2, Hnrnpu,* and *Srsf1*, suggested that noise exposure affects mitochondrial dynamics, heat-shock response, inflammation, RNA processing, and cellular stress regulation. The timepoint-based PCA also suggested that the response to noise exposure may change over time, meaning NIHL should be understood as a dynamic biological process rather than a single moment of damage.

GSE12810 served as an independent validation dataset. Because it contained only 6 samples, it did not produce strict FDR-significant genes after correction for multiple testing. However, it still provided useful validation evidence at the candidate-gene and pathway level. The strongest validation came from the targeted oxidative stress/Nrf2 analysis, which identified support for stress-response, antioxidant-defense, apoptosis-related, mitochondrial, and inflammatory genes.

This showed that even though the validation dataset was underpowered for strict genome-wide discovery, it still contributed meaningful support to the project's main biological interpretation.

The final validated oxidative stress/Nrf2-related gene set consisted of *Hspa1b, Jun, Nfe2l2, Dnm1l, Casp3, Bcl2, Dnaja1, Gclm, Nqo1, Sod2, Pink1,* and *Sirt3*. These genes represent a coordinated response involving heat-shock protection, transcriptional stress signaling, Nrf2 pathway regulation, mitochondrial fission, apoptosis, glutathione metabolism, detoxification, mitochondrial antioxidant defense, mitophagy, and redox control. This final gene set supports the conclusion that NIHL involves an interconnected network of cellular injury and protective response pathways.

The results also show the importance of being cautious with computational biology. The Random Forest model initially appeared to classify samples with 0.80 accuracy, but deeper evaluation showed that the model was affected by class imbalance. Similarly, GSE12810 did not validate every individual Dataset 1 candidate gene. These limitations do not erase the project's findings, but they show that the strongest conclusion is pathway-level rather than single-gene or diagnostic. The results should be interpreted as transcriptomic evidence of oxidative stress/Nrf2-related involvement in NIHL, not as direct proof of causation.

Genetic variation was part of the original research question and literature review, but it was not directly analyzed in the completed experimental phase. Therefore, genetic variation remains a limitation and future direction. Future studies could build on this project by connecting the final validated gene set to SNPs, GWAS results, expression quantitative trait loci, or human susceptibility studies. Larger and more balanced datasets, timepoint-specific analysis, RNA-seq validation, protein-level testing, and experimental confirmation would also strengthen the findings.

Overall, this project supports the idea that noise-induced hearing loss is not simply mechanical damage from loud sound. It is also a biological stress response shaped by oxidative damage, mitochondrial dysfunction, apoptosis, antioxidant regulation, and cellular repair pathways. By combining discovery analysis with independent validation, this project identified a final oxidative stress/Nrf2-related gene set that may help guide future research into NIHL susceptibility, prevention, and possible therapeutic targets.

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